Decentralized Crypto World Bank

A Blockchain-Based Lending Platform
with AI-Enhanced Security

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by

Md. Bokhtiar Rahman Juboraz

20301138

Md. Mahir Ahnaf Ahmed

20301083

A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

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Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

Student's Full Name & Signature:

Bokhtiar

Md. Bokhtiar Rahman Juboraz

20301138

Mahir

Md. Mahir Ahnaf Ahmed

20301083

Approval

The pre-thesis 1 report of final year project titled "Decentralized Crypto World Bank: A Blockchain-Based Lending Platform with AI-Enhanced Security" submitted by

1. Md. Bokhtiar Rahman Juboraz (20301138)

2. Md. Mahir Ahnaf Ahmed (20301083)

Of Spring, 2026 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science on February 20, 2026.

Examining Committee:

Supervisor:
(Member)

Mr. Annajiat Alim Rasel
Senior Lecturer
Department of Computer Science and Engineering
BRAC University

Project Coordinator:
(Member)

Dr. Md. Golam Rabiul Alam
Professor
Department of Computer Science and Engineering
BRAC University

Head of Department:
(Chair)

Dr. Sadia Hamid Kazi
Chairperson and Associate Professor
Department of Computer Science and Engineering
BRAC University

Ethics Statement

This project is designed to be functional on public blockchain testnets, with no intention for monetary profit, nor to create institutional leverage over decentralized financing. No personal banking data, or private financial data, or codes of existing projects, are used in the development of the prototype platform and training AI/ML data. The full project prototype build is to be demonstrated in the final defence of the project; the platform is expected to be ready for testing and troubleshooting by pre-thesis 2. All the development workflow can be traced by the official GitHub repository for the project, and the planning, coding, testing, and usage of different dependencies for this project is considered open-source with MIT licensing, and will be open for contribution after the submission of the final build. For the training of the AI models, AI-assisted financial decision-making, risk assessment, privacy features, on-chain and off-chain data processing and data storing, we acknowledge the ethical considerations while making decisions for the development of the project.

Abstract

In the modern world, global financing relies on institutional structures that are multilayered and complex that extend throughout the borders of nations, adapting complex protocols to maintain balance in the economy; however, these systems often struggle to maintain transparency and equal access to credit across the world. Decentralized financing has improved transparency in financing by adapting programmable infrastructures that can automate exchanges or structured governance, but the existing models use mostly flat architecture that does not match with the traditional institutional hierarchies or governance that is well structured based on global economic contexts. This creates a scope to study and explore how tier-based financial systems that mimic the functionalities, but in a decentralized environment with the flexibility and adaptability. As an attempt to address this opportunity and to demonstrate what is possible, we present our research-based project *Crypto World Bank*, a formally specified prototype which is based on blockchain-based institutional finance, that includes coordinating capital allocation across multi-tiered hierarchical governance structures. The primary focus of the project is a smart-contract that includes a hierarchy with a pipeline of oracle-based risk-analytics (Random Forest, Isolation Forest, SHAP, which will be trained and benchmarked with existing similar systems); an optional intelligent MCP agent as a feature to interact with the UI with more comfort, while all the core banking systems remain intact by using wallets. The banking architecture includes risk intermediation, liquidity management, deposit mobilization, credit allocation, payment settlement, and other adjacent financial services. The development workflow including frontend, smart contract, oracle integration and overall system architecture and design is specified in this paper; implementation and testnet deployment are planned across four phases in the final stages. The goal of this project paper is to illustrate how a hybrid system of design of decentralized system can converge with institutional systems, to set a foundation for research on how a hierarchically governed crypto financial system can be developed.

Keywords: Blockchain, Decentralized Finance, Deposit Mobilization, Explainable AI, Financial Sustainability, Group Lending, Hierarchical Lending, Institutional DeFi, Multi-Entity Co-Lending, Oracle-Mediated Risk Analytics, Reserve Management, Smart Contracts

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

Acknowledgment

We would like to express our sincere gratitude to our panel members for their guidance and support throughout this project. We also thank our supervisor, Mr. Annajiat Alim Rasel, Senior Lecturer at the Department of Computer Science and Engineering, BRAC University, for his guidance and support. Finally, we thank the Department of Computer Science and Engineering at BRAC University for providing us with the resources and academic environment to pursue this work.

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Chapter 1

Introduction

The *Crypto World Bank* is a research-based project for exploring how blockchain technology can be shaped to fit into institutional financial systems with all the added benefits that come with a decentralized financing system. Based on our findings that had similar motivation to our study, existing DeFi protocols show isolated functionalities, such as Aave for lending, Uniswap for exchange, and MakerDAO for stablecoins. This project specifies a tier-based, hierarchically governed platform that includes a mixture of institutional financing with DeFi, accommodating deposit mobilization, credit allocation, interbank liquidity, installment repayment, FX facilitation, solidarity group lending, reserve enforcement, syndicated co-lending, and risk-based governance covering all four tiers that systematically mirrors real-world institutional financing. At the top, Tier 1 represents the World Bank reserve; the core banking hierarchy is specified in Solidity and being improved until final deployment. Another significant feature—the multi-entity contract suite—is fully specified in Chapter 3, which is scheduled for practical development in Phases II–III.

The system is designed for three categories of participant: a programmable World Bank reserve as Tier 1, national and local development banks serving in Tiers 2–3, and retail clients in Tier 4. Retail clients get access to savings, group loans, and installment lending features. This hierarchical structure in this project was motivated by unbanked and MSME financing gaps [9, 11]. The project is not built with any motivation to deploy only in regions with access to advanced technologies; rather, proper distribution of access to finance and fairness is the leading cause of this attempt to find suitable ground to adapt DeFi among the general population. The whole project is open source and available in our GitHub repository. The added benefits of blockchain: it provides state visibility, tamper-evident audit trails, and programmable rule enforcement among institutions having competition, as demonstrated by the renowned World Bank FundsChain programme [27] and JPMorgan Kinexys [28]. From the study of institutions, open ledger infrastructure and institutional hierarchy are not mutually exclusive.

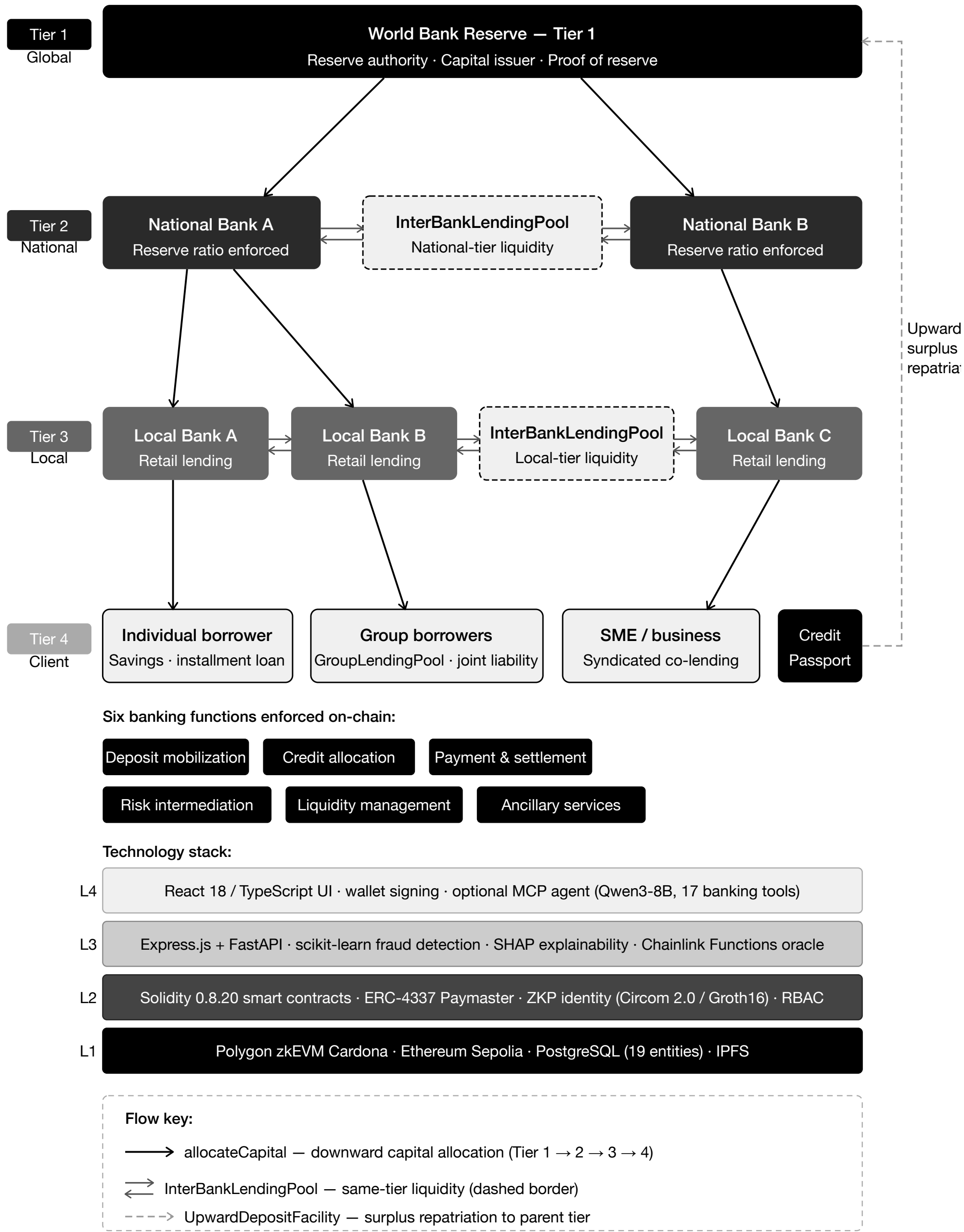


Figure 1.1: Crypto World Bank system overview

1.1 Background

Local lenders, or our target users, for this aspect of lending through layered institutions—supranational bodies that is provided by development finance channels have structural costs. Idle capital is usually locked by correspondent banking in pre-funded nostro/vostro accounts [14], cross-border transactions that take around two to five days cost roughly $45 per transaction [15], charges for remittance corridors average 6.49%, which is above the UN SDG target of 3% [16]. Due to unequal access to traditional banking systems, 1.4 billion adults remain unbanked [9] and MSMEs face a $4.5 trillion annual financing gap [11].

In decentralized financing, lending, collateral, and interest can be automated by enabling rules on-chain transparently. However, institutions like Aave v3 that holds $26.3 billion TVL [17] and the broader DeFi lending market exceeds $55 billion [8], use flat pool design that ignore institutional hierarchy, inter-tier exchange, and accessing clients via local bank lend path which is more generalized for normal people with less knowledge about DeFi.

Contribution of smart-contract and blockchain. Smart-contracts can develop lending rules as immutable, atomic, with auditable on-chain logic. This technology replaces the bilateral ledger reconciliation with a system of shared record, when a local bank initiates a loan, the World Bank's reserve ratio updates on the same transaction status [27]. DeFi extends this feature to financial services without licensed custodians. Here in The *Crypto World Bank* adds three dimensional feature that is missing from the existing institutional protocols: institutional hierarchy (four-tier capital flows), multi-entity operations (solidarity groups, syndicated loans, tranched pools), and a connection to the evolving development finance framework identified as absent in the literature [1].

Optional conversational interface. Every user interacts with this system primarily through the React UI and wallet credential signing. For development purposes and demonstration, a locally hosted agent (Qwen3-8B + MCP, 17 banking tools) provides an alternate path to get the essential tasks done, including all the customer services that require system interaction information guidance and assistance in acquiring a loan or exchange finances with the system. In the final build it is planned to be replaced with secured API-based services.

Technical foundations. **Consensus.** The chain is an append-only ledger: nodes agree on history, and no one can rewrite past blocks without controlling most of the network [50].

EVM execution [41]. Loan logic runs as deterministic bytecode on Ethereum-compatible chains. Storage writes are the main cost driver (Section 5.1).

State machines. Each loan moves through clear stages—`PENDING` → `APPROVED` → `ACTIVE` → `REPAYING` → `COMPLETED/DEFAULTED`—with only authorised roles allowed to advance it.

Oracles [45, 76]. Contracts cannot read off-chain data on their own. ML risk scores reach the chain via a commit-reveal oracle (Section 3.2); Chainlink Feeds, Automation, and Proof of Reserve handle prices, due-date checks, and reserve attestation. Live ML wiring starts after pre-thesis 2.

Polygon zkEVM. We deploy on Cardona testnet (Section 3.2) for low fees and L1-backed validity proofs.

1.2 Rationale and Motivation

Real development finance depends on layered institutions, yet settlement friction, opaque reserves, and correspondent-banking idle capital leave both intermediaries and unbanked clients underserved [16, 90]. Flat DeFi protocols automate lending at scale but ignore institutional hierarchy, cross-tier flows, and the Local-Bank-to-client path that microfinance and development banks use in practice [1]. This study is motivated by the opportunity to encode that hierarchy on a programmable ledger—with enforced reserves, role-based governance, and auditable capital movement—while supporting (not replacing) human credit judgment through lightweight analytics and an optional conversational interface. The thesis specifies a reusable architecture; it does not claim a particular national deployment target.

1.3 Problem Statement

Three primary forces of motivation for this study: development-finance has restricting transparency and settlement; the absence of multi-tiered banking functions in DeFi (deposits, solidarity and lending); the keen interest in combining blockchain auditability with ML computational analytics. There are six structural inefficiencies noticeable in development finance routes for individual borrowers from national and local banks:

1. **Lack of transparency:** there is very little chance of the lower tier users to verify capital management, reserve levels or lending decisions are made by the higher tiers, very less transparency on how money flows through the different levels of systems. Only the top tiers get the information and records of reserves and audited at most quarterly.
2. **Sluggish settlements and idle capital:** cross-border transactions are quite expensive, average $45 per transaction [15], and time consuming taking two to five days [21].
3. **Inconsistent risk evaluation:** fraud and credit assessment are completely relied on human judgement, borrowers are generally re-evaluated at each tier or institutions based on information available at that tier, no unified credit history.
4. **Access and trust barriers:** the cost of inter-institutional trust is quite expensive for legal and audit process. It creates congestions in developing countries where investors earn returns roughly 3% below compared to developed markets [18].
5. **No programmable savings:** depositors in developing economies have no real-time access to visibility into how savings are processed and deployed.
6. **No on-chain group credit:** without individual collateral, borrowers cannot access programmable on-chain mutual group lending with shared responsibility. Real-world organizations like BRAC exist but does not function like DeFi.

DeFi enabled automatic lending transparency at scale such as Aave v3 with $26.3 billion TVL, $17.7 billion borrows [17], Compound v3 ($1.4 billion) [19], MakerDAO with $10.5 billion stablecoin issuance [20]. Despite large market coverage they exhibit four limitations for institutional development finance:

- **Flat lending models.** Aave/Compound uses one protocol for many asset pools (USDC, ETH, etc.). There is no cross-tier allocation, no institutional hierarchy, no difference in role-based governance. Here our CWB models institutional relationships that go deep into what kind of services to provide to what kind of roles/institutions to increase liquidity in overall lending and exchange situations.
- **No AI risk analytics.** DeFi relies on collateral ratio, utilization curves. There is no behavioral fraud detection or explainable credit support system.
- **No tiered governance.** DeFi protocol treats tokens as a measurement of voting, not based on what is the role of the client or user or institution in the bigger to smaller picture as it lacks relation to a deeper level like institutions.
- **Single-level institutional DeFi.** Maple & Goldfinch [51] with $2.6–3.8 billion TVL and $680 million originations in 18+ countries respectively serve institutions, but

none models a full multi-tier banking system with interbank flows, lack hierarchical capital flows.

1.4 Objectives

The objectives of this project are:

1. Design and specify all features and constraints of a four-tier lending architecture on an EVM-compatible blockchain and complete a full testnet deployment (World Bank → National Bank → Local Bank → Client). The whole development cycle is planned for Phases I–IV.
2. Design not only pool lending, but specify an extended banking product suite on the hierarchical system (deposits, savings, and solidarity group lending).
3. Investigate a lightweight off-chain analytics layer using ML, training and testing the models in practical testnet for fraud detection, anomaly detection, explainable review, and an optional MCP-based conversational interactive agent for ease of use.
4. Specify programmable on-chain lending with borrowing limits based on financial behaviour analysis, installments, and role-based access control and information via smart contracts.
5. Plan and execute testnet deployment using Polygon zkEVM Cardona, Ethereum Sepolia, and evaluate hierarchical controls in the final thesis phase.
6. Document governance for future improvements, security analysis and feature improvements, role-based regulatory considerations, and a controlled rollout toward sandbox piloting.

1.5 Proposed Solution

The Crypto World Bank will implement a four-tier on-chain lending model: Tier 1 World Bank reserve custodian → Tier 2 National Banks → Tier 3 Local Banks processing retail loans → Tier 4 clients building on-chain credit history (Figure 1.1), with six standard banking functions enforced on-chain—deposit mobilization, credit allocation, payment and settlement, risk intermediation, liquidity management, and ancillary services—rather than discretionary audit alone. Banks also borrow from peers, park surplus with parent tiers, and syndicate large loans; the platform replicates four-tier downward flow, same-tier pools, upward deposits, and multi-bank co-funding on-chain (Figure 1.2).

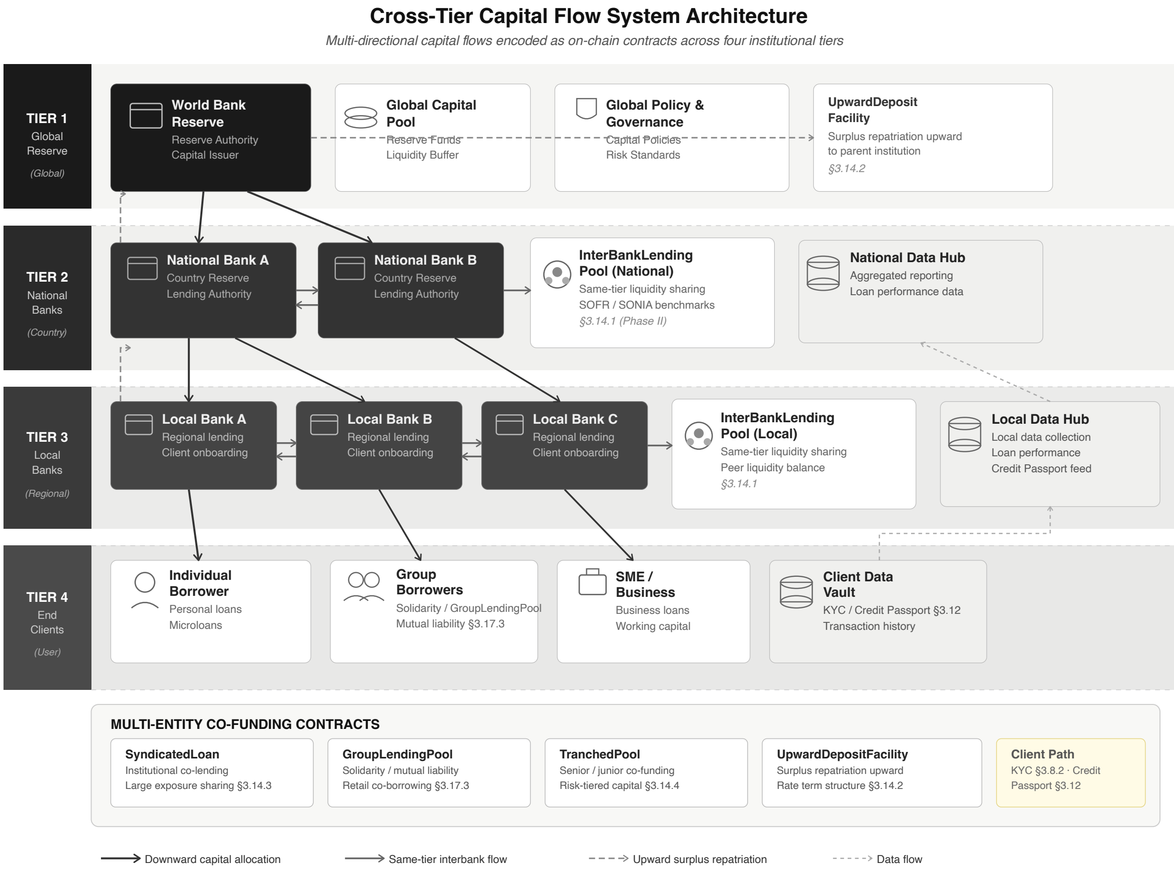


Figure 1.2: Cross-tier lending flow directions

Can a group of entities lend or fund together? Syndicated loans, solidarity groups, tranched pools, and upward surplus deposits (Figure 1.3; Section 3.2) coexist with same-tier peer liquidity pools (Section 3.2); retail clients enter through Local Banks via KYC, Credit Passport, and borrowing limits, with syndication reserved for large loans.

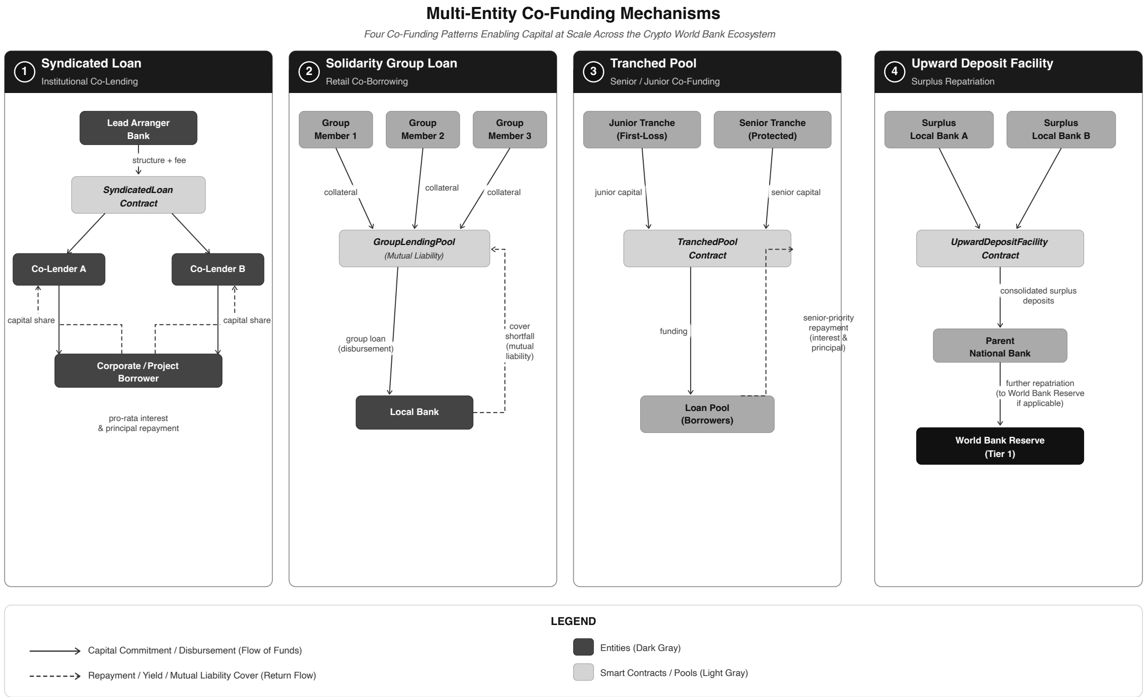


Figure 1.3: Multi-entity co-funding mechanisms: syndicated loan, solidarity group, tranched pool, and upward deposit

1.6 Methodology in Brief

There are four Agile phases to this methodology for this project. This report is designed with the architecture specification. The development phase will start from Phase II after pre-thesis 1. In this report there is architecture, contracts, database, and ML pipeline. The Phase II development phase will demonstrate the loan cycle. Phase III will start the implementation of ML scoring and the Chainlink oracle (Section 3.2). Phase IV will show the evaluation, metrics, and the overall feasibility of the project in the real world. Stack: Solidity, React, Express/FastAPI, scikit-learn on Polygon zkEVM Cardona. There is also an AI chat agent that will be built as an optional feature that will run alongside the project to help the clients navigate better with the system.

1.7 Scopes and Challenges

In scope now. In our development phases, we have calculated that we can implement a limited amount of our full vision: the four-tier lending architecture, loan lifecycle workflow, borrowing limits for the clients and the banks, the design of Oracle fraud analytics, and the training and the live demonstration of the AI/ML workflow. It can be shown to a limited amount on the prototype builds only.

Out of scope for now. Mainnet deployment, live retail payments in the real world, production KYC/AML at the banking level and client interaction, stress testing the whole project, and providing an MVT checklist before the product is ready is not quite possible yet.

Main risks. There are a couple of risks that we have noticed for our project, starting with the fraud labels may not be completely accurate based on the limited amount of data we have for our system architecture. We need more synthetic data that will be collected over time. If regulations have changed—mitigated by testnet-only builds—then we might have to change the architectural elements. The gas toll on micro-loans may not be sustainable. The ML layer with SHAP may keep borrowers in the loop in the system since it is not 100% accurate and accuracy of the system depends on the training process. Rest of the economics and trade-offs are discussed in Chapter 5.

Economic sustainability. On Cardona, 10–15 txs on a typical loan cost under $0.15—small next to interest earned. Mainnet micro-loans would need tighter gas work.

Institutional capture. Another huge issue that needs to be addressed is that Tier 1 and the National Banks, which have huge reserves and influence over the rates, reserves, and who gets credit, might influence the overall system and cost-to-performance ratio. It might temper the trust and relationship with the retail customers and smaller banks.

Stablecoin-first lending. Retail clients are not to be carrying high-priced positions like ETH, since the prices of Ethereum might fluctuate. MiCA and the U.S. GENIUS Act support the idea that stablecoins should be more generalized for authorized cryptocurrency. There are many settlement projects like *mBridge* and *Agora* [64, 65], moving the value of cryptocurrencies across borders, but it is not a replacement for our lending system. The architectural system for stablecoin pools and L2 deployment (Section 3.2) gives the liquidity move fluidly from the top tier to the bottom, and the surplus to go up from the bottom.

1.8 Thesis Organization

- **Ch. 1 — Introduction:** problem, objectives, approach (**CO1**).
- **Ch. 2 — Literature:** PRISMA review and protocol comparison (**CO9**).
- **Ch. 3 — Architecture:** tiers, contracts, data model, identity, products, security (**CO2–4**, **CO12**).
- **Ch. 4 — Methodology:** build phases, MVT scope, ML oracle, optional agent (**CO5**, **CO10**).
- **Ch. 5 — Evaluation:** feasibility, benchmarks, limits (**CO6–7**, **CO11**).
- **Ch. 6 — Conclusion:** contributions and future work; appendices hold contracts and deployment notes.

Chapter 2

Literature Review

2.1 Preliminaries

Important terms—XAI, PoS, CBDC, solidarity/group lending, DeFi, smart contracts, over-collateralization, ALM, flash loans, ZKP, and Health Factor—have been introduced in Glossary appendix. This section has been expanded in appendix D that will be presented in the final versions of the complete project file.

2.1.1 Review Methodology (PRISMA-Style Frame)

We did our research to find the appropriate papers and highly qualified papers that goes without project from ACM, IEEE, Elsevier, MDPI, arXiv, and BIS/World Bank (2017–2026) with PRISMA-style screening [52]. Primary topics for this search: DeFi lending, fraud ML, tiered finance, and smart-contract security.

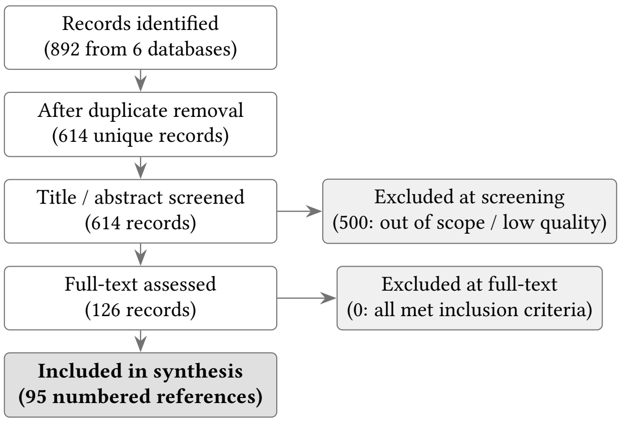


Figure 2.1: PRISMA-style literature screening flow

2.2 Review of Existing Research

We explain our literature review, and our findings from the papers and the work of the authors, in the subsections shown below.

2.2.1 Decentralized Lending and DeFi Protocol Design

For a big picture, Werner et al. [1] show that DeFi lenders use flat pools with no bank-style tiers; we have found a gap that can be filled by introducing our project. According to Bastankhah et al. [2], adaptive borrow rates work better than Aave-style curves for financial flows, which we introduced in our kinked-rate module. Dao et al. [43] show how on-chain limits can get off-chain credit scores that in general are applied at our Local Bank level.

2.2.2 Machine Learning, Explainable AI, and Extended AI Security

For fraud detection, Palaiokrassas et al. [3] engineer behavioral features that do a really good job in DeFi transaction fraud detection, not just hashes. We plan to use Random Forest in Phase III that will further strengthen the fraud detection level with the mixture of hash functions that is built into the DeFi protocol. Moreover, for the situations when the fraud detection level is scarce, we will use Liu et al. [6] Isolation Forest. We also studied a very in-depth comparison of SHAP versus LIME by Adom et al. [4], which shows a significant advantage by using SHAP for loan detection; approvers in bank and authority level will see SHAP screening results before signing.

2.2.3 Hierarchical Distribution, Cross-Tier Flows, and Group Lending

Tan [5] demonstrated how CBDC flows in bank-to-bank and bank-to-client financial transactions. We extended this idea in our project by showing tier-based reserve checks. For layered reserve contracts, *Project Pine* [79] has shown a prototype that is similar to our `InterBankLendingPool` and the syndicated-style loan flow. We also integrated the style of microcredit admin cost with smart contracts based on the study of Asaduzzaman et al. [38]. The structure of `GroupLendingPool` function is the on-chain version, without claiming that we are trying to replace BRAC-style social microfinancing enforcement.

2.2.4 Smart Contract Security and Layer 2 Scalability

To improve the security system of the project, we are following Atzei et al. [7], which lists the main Ethereum contract attacks. Since the tier-based system is very complex, many transactions on Cardona might be triggered. Gas prices can be cut to 30–50% by transaction batching [13], with further DeFi gas optimisation [39].

2.2.5 Monetary Policy, Tokenization, and Institutional Landscape

Very large institutions like World Bank FundsChain [27] and similar institutions show that development finance is already moving on-chain for auditability; we are further improving this by encoding rules, not just managing disbursement logs.

2.2.6 Graph ML, Federated Learning, Regulation, and Identity Primitives

We noticed the usability and rise of stablecoin and went further deep into it and try to explore how stablecoin pools for retail clients can be strengthened by using MiCA and the U.S. GENIUS Act [61, 63]. For our project, soulbound tokens will be used that fit into our Credit Passport idea, which was extended by the idea of Buterin et al. [55]. This introduces a secured identity on-chain, a portable reputation, without the fear of exposing personal data.

2.2.7 Oracle-Mediated ML and Optional LLM Interfaces

Lastly, for an optional feature to our project, future studies can be made and features can be added to the system. Caldarelli [76] describes how ML scores still have some issues that go beyond price oracles; we focus on this part and extend our commit-reveal or Chainlink path targets. We collected some reputable databases, and more to be added such as the BCCC fraud dataset [71] that goes with our system. This section will be continued and extended after pre-thesis 2.

Table 2.1: Literature review summary (Part 1 of 2)

Author & Focus	Methodology	What they found & how we use it
Palaiokrassas et al. (2024) [3] · DeFi fraud	XGBoost, NN on 54M+ txs	We plan to use behavioural features in Phase II for Random Forest, as it lifted F1 to 0.76–0.85
Adom et al. (2025) [4] · XAI lending	LIME / SHAP comparison	Approvers get a SHAP brief before signing, as it gives a clearer result than LIME on loan data
Tan (2023) [5] · CBDC hierarchy	Developing-nation model	CBDC backs the four-tier layout and works as a bank-to-user tier
Liu et al. (2008) [6] · Anomaly detection	Isolation Forest	Flags odd wallets without many labels; backup when fraud data is thin
Atzei et al. (2017) [7] · SC security	SoK of Ethereum attacks	Lists re-entrancy and access-control risks; guides OpenZeppelin guards
Werner et al. (2022) [1] · DeFi SoK	Survey of 12+ protocol types	Main gap CWB targets is DeFi lending having no bank hierarchy
Bastankhah et al. (2024) [2] · Adaptive lending	Dual fast/slow control	Dynamic rates support our kinked-rate modules as it beats static curves
Hartmann & Hasan (2021) [32] · P2P lending	Smart-contract origination	Gas is cut in on-chain P2P, which is a minor input for loan-facing clients
Hasan et al. (2022) [33] · Federated fraud	FL on encrypted features	Lack of raw-data sharing features by banks leaves future cross-bank options
Wang et al. (2020) [34] · SC vulnerabilities	ML on bytecode	Common bytecode bugs spotting by ML helps provide extra security over manual audit
Liao et al. (2019) [35] · Audit automation	Classification + fuzzing	Workflow idea for Phase V is inspired from ML prioritizing fuzzing speed

Table 2.2: Literature review summary (Part 2 of 2)

Author & Focus	Methodology	What they found & how we use it
BIS et al. (2020) [37] · CBDC design	Retail distribution analysis	Standard CBDC two-tier is extended to four-tier with on-chain reserves
Kiff et al. (2020) [82] · CBDC models	IMF architecture survey	Matching National/Local roles; retail CBDC flow implemented through supervised banks
Bindseil (2020) [81] · CBDC rates	ECB tiered remuneration	Upward deposit spread idea inspired from tiered rates that protect bank intermediation
Dong et al. (2023) [80] · Supply-chain finance	Three-tier game model	Capital can move down a chain, parallel to World Bank → client allocation
Project Pine (2025) [79] · Hierarchical SCs	NY Fed / BIS prototype	IBLP and syndicated loans used on-chain layered reserve contracts prototype
CPMI/FSB [14, 21] · Cross-border settlement	Nostro/vostro cost analysis	Tiered on-chain flow is better than legacy settlement as it is slow and costly
World Bank (2025) [27] · Dev. finance blockchain	13-project pilot	Programmable lending rules added with dev-banks system
Chen et al. (2024) [30] · Multi-chain lending	Cross-chain design	Scaling by splitting loads throughout chain is later option, not for Phase I
Yang & Cui (2023) [31] · Protocol evaluation	Quantitative risk metrics	Gives utilisation and liquidation benchmarks for health checks tier
Asaduzzaman et al. (2020) [38] · Smart micro-credit	Smart-contract MFI	Model for `GroupLendingPool` is smart contracts, which cuts the admin costs in Bangladesh
Wang et al. (2021) [13] · Tx batching	iBatch middleware	Per call saved 15–59% in gas price by which we deploy on Cardona L2
Kim & Kim (2024) [39] · Gas optimisation	DeFi parameter tuning	∼18% gas cut on Ethereum pools helps our L2 deployment
Yuan (2024) [40] · SHAP credit default	RF, GBM + SHAP	Deploying on Cardona L2 as batching saved 30–50% gas; SHAP is also used for our decline path

Feature	Aave	Comp.	Maker	Maple	Gfinch	CWB
Institutional hierarchy	✗	✗	✗	✗	✗	●
Cross-tier capital flow	✗	✗	✗	✗	✗	⧗
Same-tier interbank lending	✗	✗	✗	✗	✗	⧗
Syndicated / co-lending	✗	✗	✗	Part.	Part.	⧗
Upward capital repatriation	✗	✗	✗	✗	✗	⧗
Solidarity group lending	✗	✗	✗	✗	Part.	●
AI/ML fraud detection	✗	✗	✗	Man.	Man.	●
SHAP explainability	✗	✗	✗	✗	✗	●
ZKP KYC compliance	✗	✗	✗	✗	✗	●
Kinked interest rate curve	✓	✓	Gov.	✓	✗	⧗
Role-based access control	Part.	Part.	Gov.	✓	Part.	●
Institutional / L2 focus	✗	✗	✗	✓	Part.	●
TVL / Status (2026)	$26B	$1.4B	$10B	$2.6B	$680M	Planning

Table 2.3: Protocol comparison on eleven architectural dimensions

2.3 Summary of Key Findings

Four points stand out after reviewing and Table 2.3:

- **Flat DeFi vs. tiers.** Large pool lenders [1, 8] have no hierarchical bank currently, and it opens a door to explore how a hierarchical system would behave with DeFi. CBDC and Project Pine work [5, 79] prove that nobody combines a four reserve-enforced tier with group lending and cross-tier financial flows and an ML oracle in a singular architecture.
- **ML and security.** ML and security layers combined with behavioural fraud features and SHAP [3, 4, 6] further extend the usability, feasibility, and adaptability of our project. Based on these literature reviews, we further improve our Phase III pipeline; contract attack catalogues [7] and L2 gas batching [13, 39] justify OpenZeppelin guards to be added to our project.
- **Groups and institutions.** We explored smart-contract microcredit in Bangladesh [38] and World Bank on-chain pilots [27] that gave us the idea and improved the implementation of `GroupLendingPool` function and institutional lending system.
- **Regulation.** For regulation improvement and securities, MiCA and GENIUS [61, 63] are accepted by stablecoin-first pools. Soulbound credit tokens [55] fit the Credit Passport without putting KYC on-chain.

Chapter 3

System Architecture and Design

3.1 Prototype Scope and Stack

Table 3.1: Prototype scope (summary)

Feature	Status
Four-tier RBAC (World / National / Local / Client)	Specified (Phase I)
Loan lifecycle + installments	Specified / Phase II
SavingsVault, GroupLendingPool, IBLP	Specified / Phase II
RF + IF + SHAP oracle pipeline	Planned (Phase III)
ZKP KYC, testnet deploy, Foundry sim	Planned (Phase IV)
Optional MCP agent (17 tools)	Optional (Phase III–IV)

3.2 System Architecture

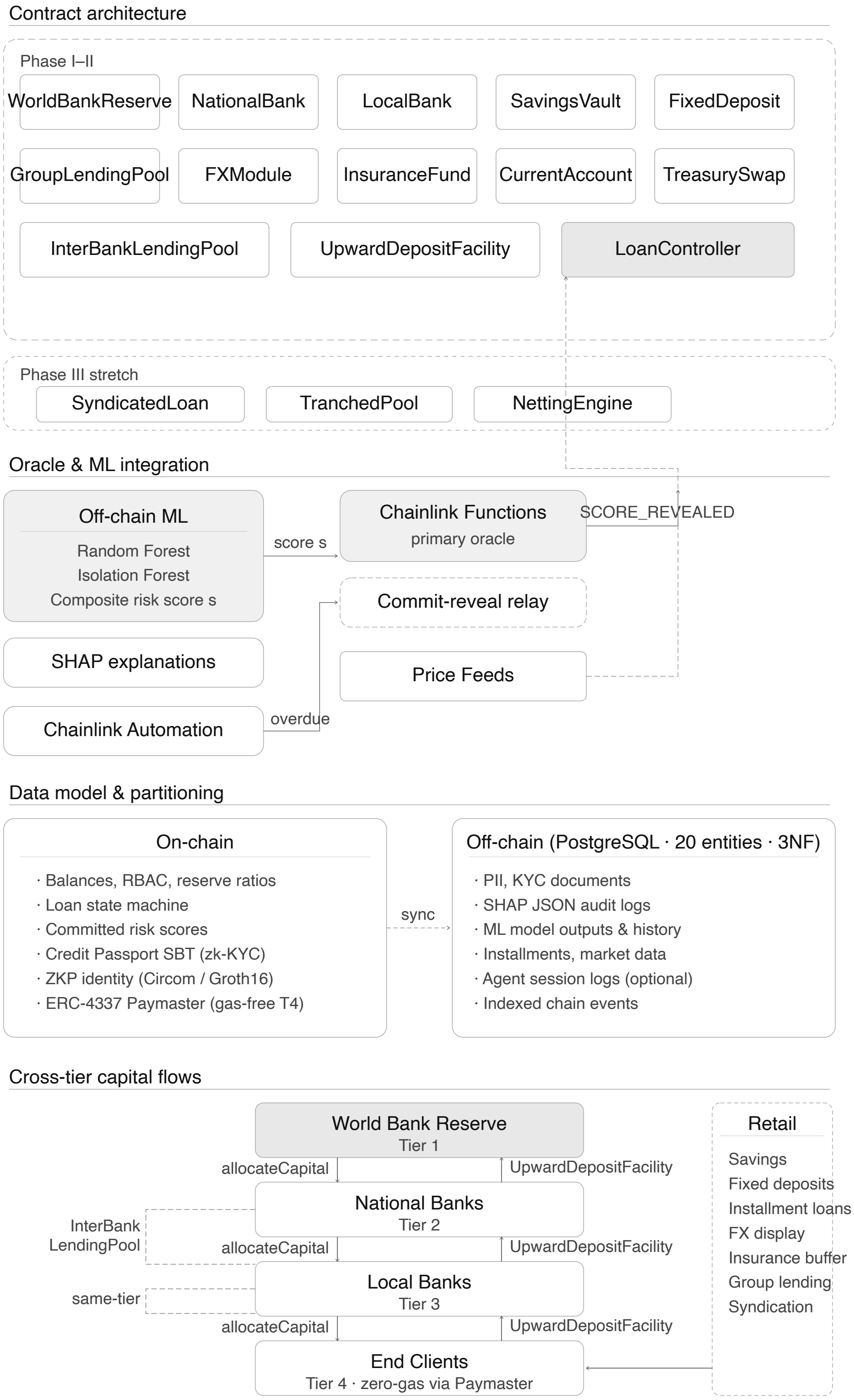


Figure 3.1: CWB system architecture — contracts, oracle/ML integration, data partitioning, and cross-tier operations

3.3 Architecture Diagrams (Full Set)

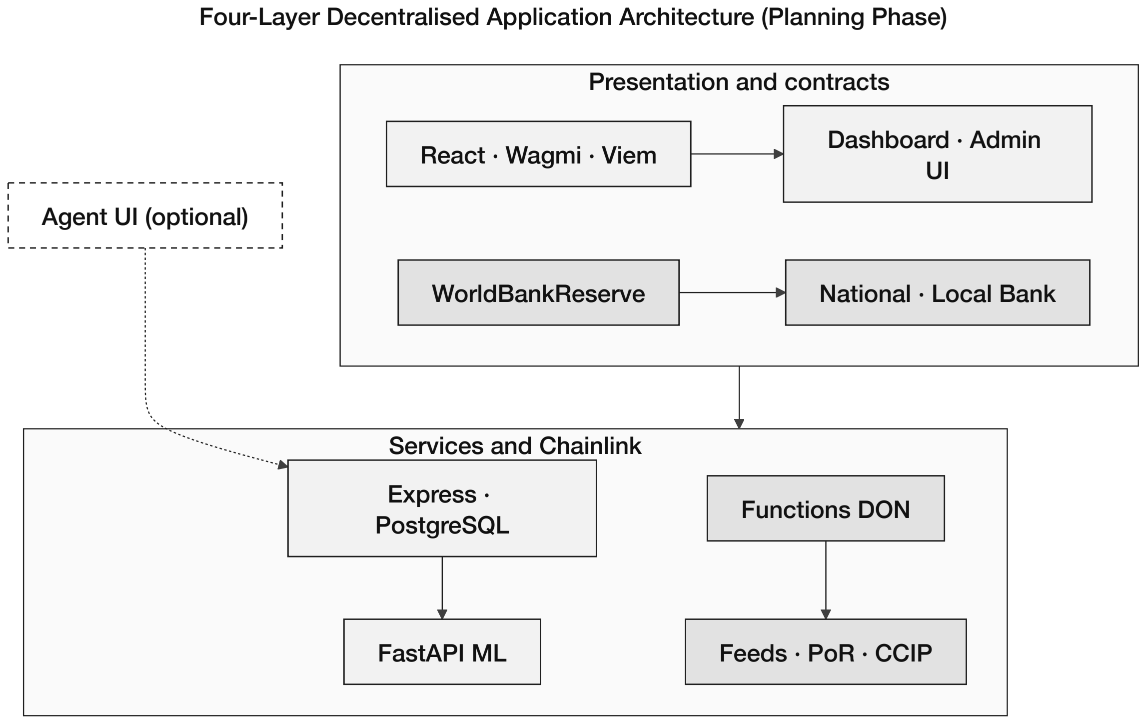


Figure 3.2: Four-layer application architecture

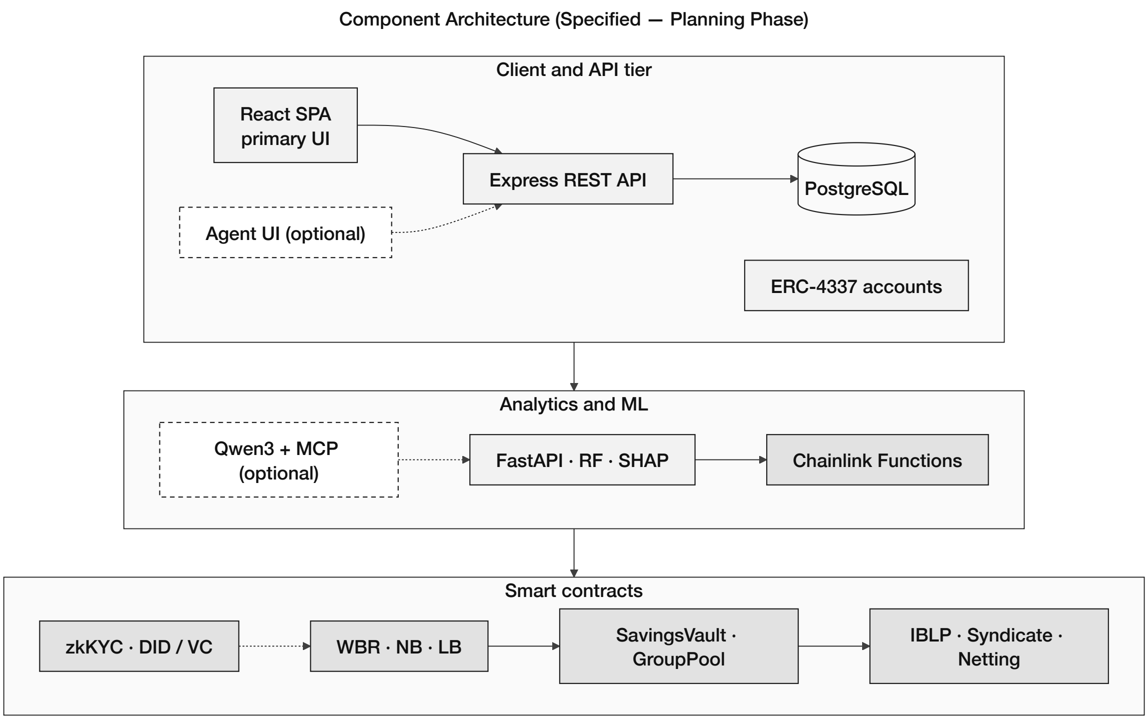


Figure 3.3: Component diagram (planning phase)

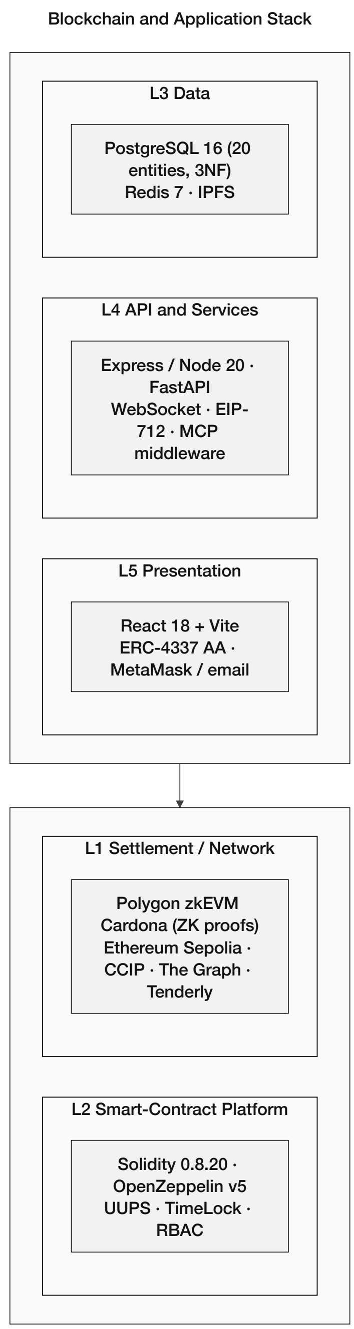


Figure 3.4: Layered blockchain and application stack

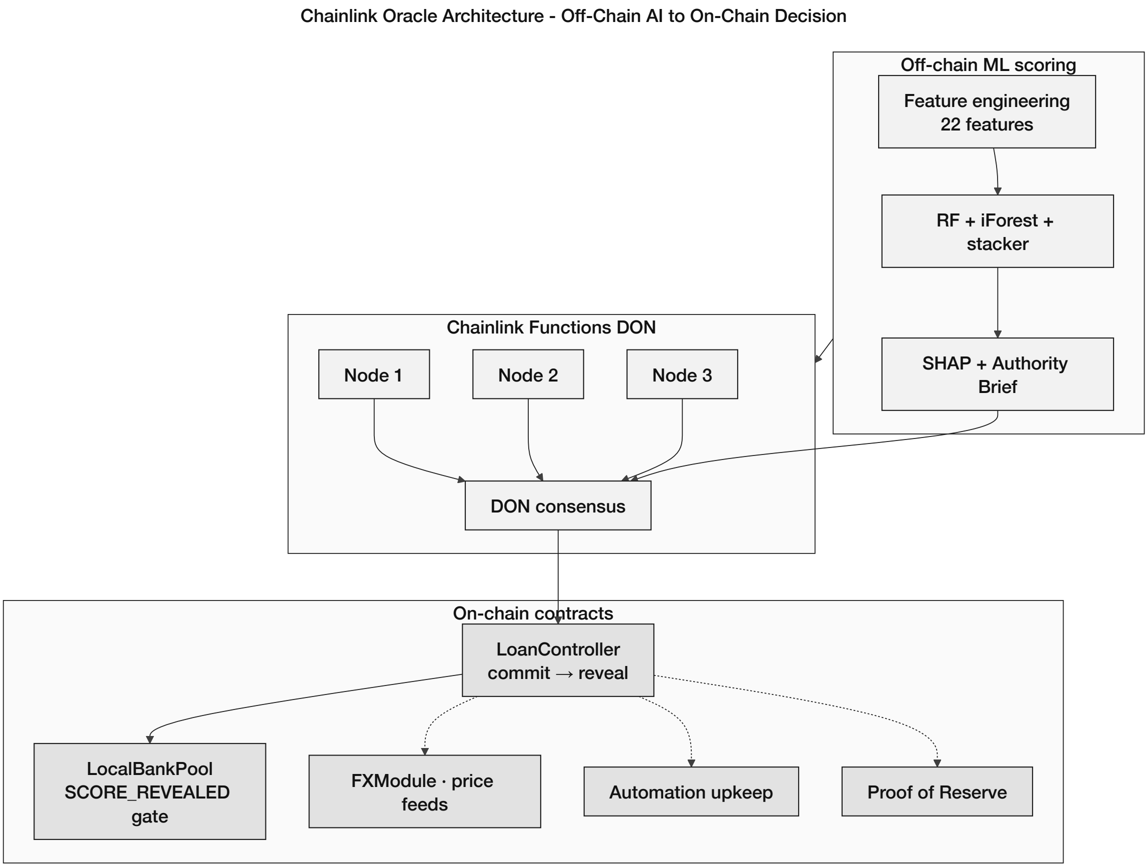


Figure 3.5: Chainlink oracle architecture

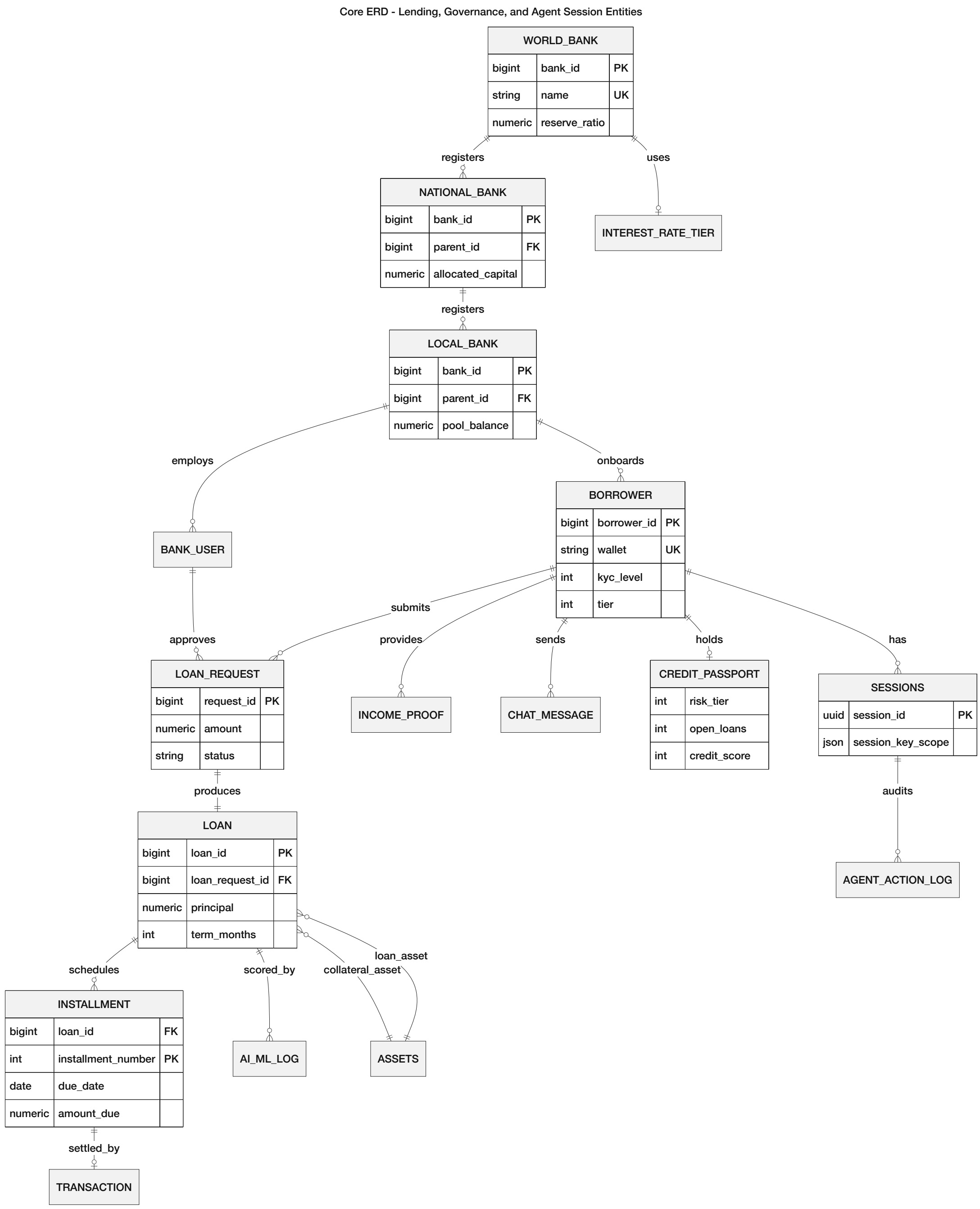


Figure 3.6: Core system entity graph

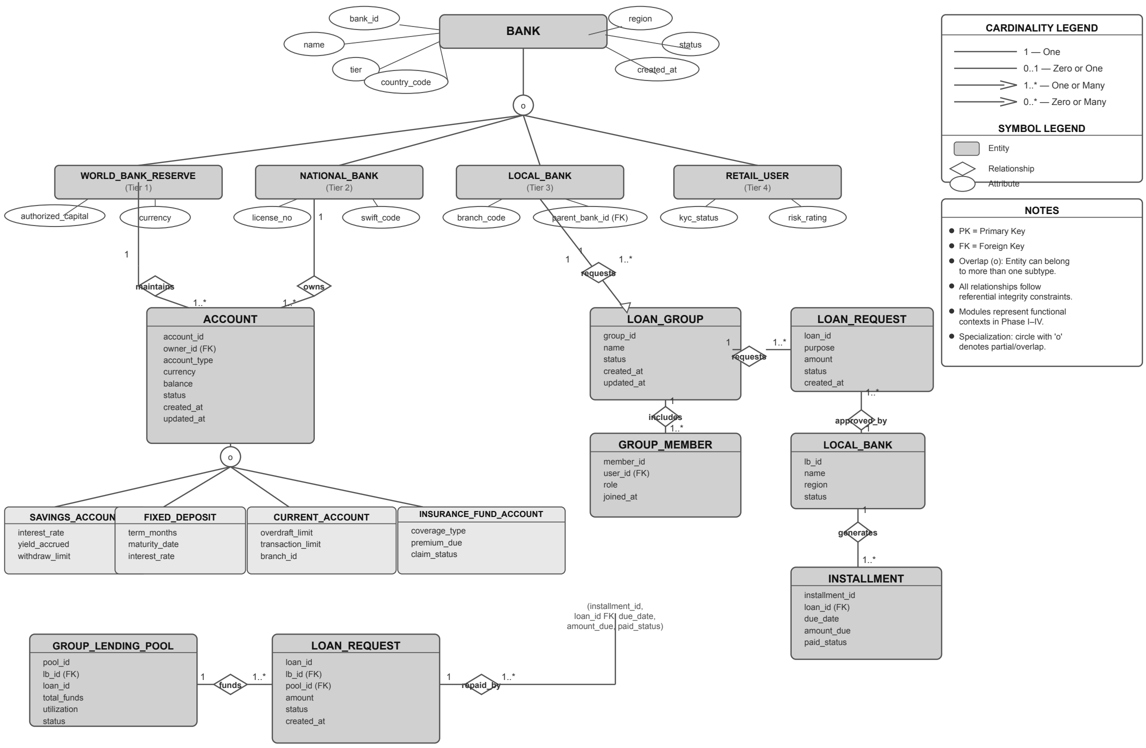


Figure 3.7: Enhanced EER model

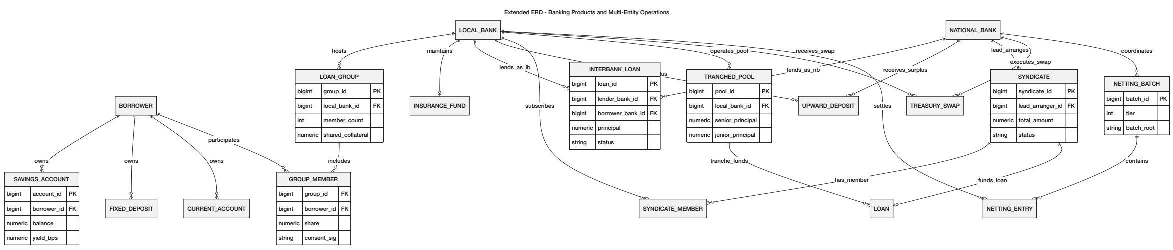


Figure 3.8: Extended ERD entities

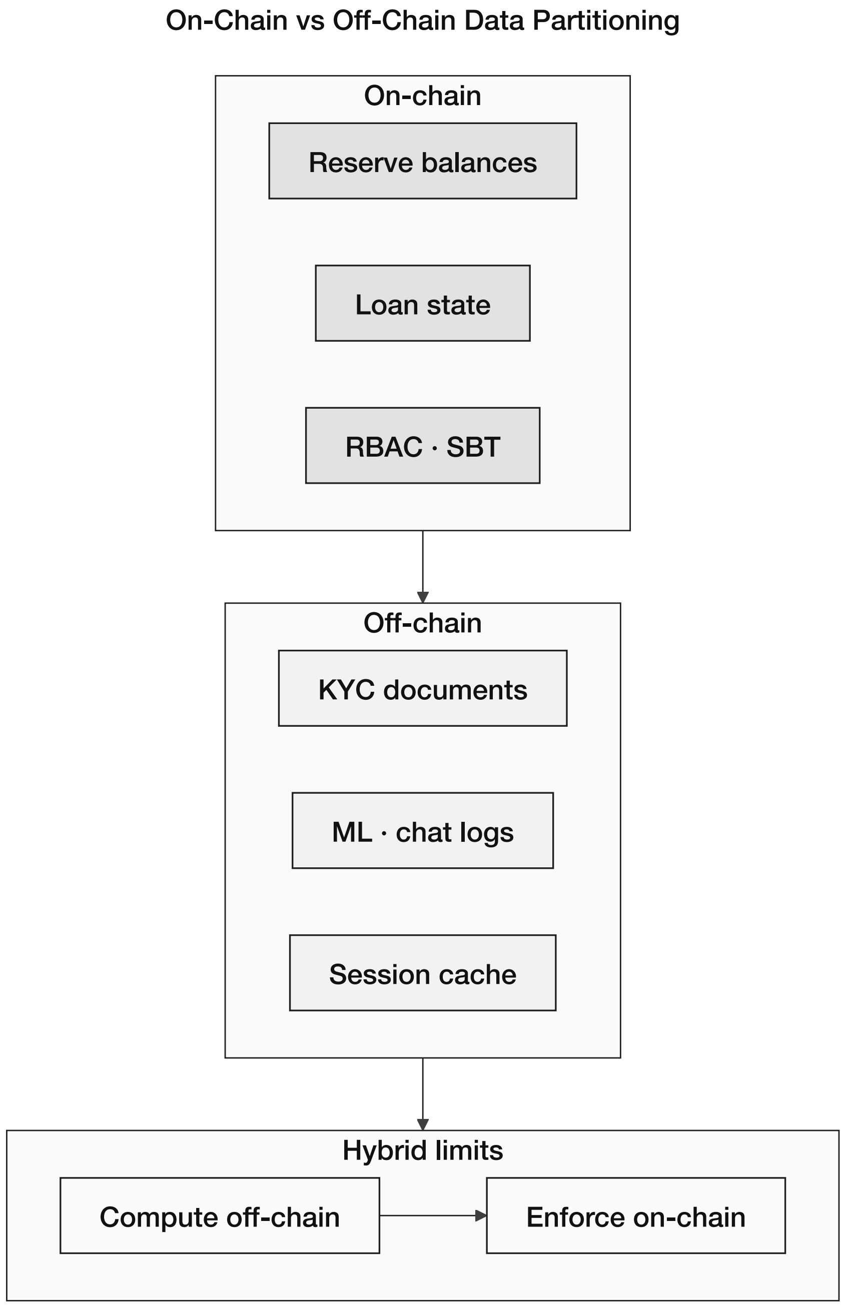


Figure 3.9: On-chain versus off-chain data partitioning

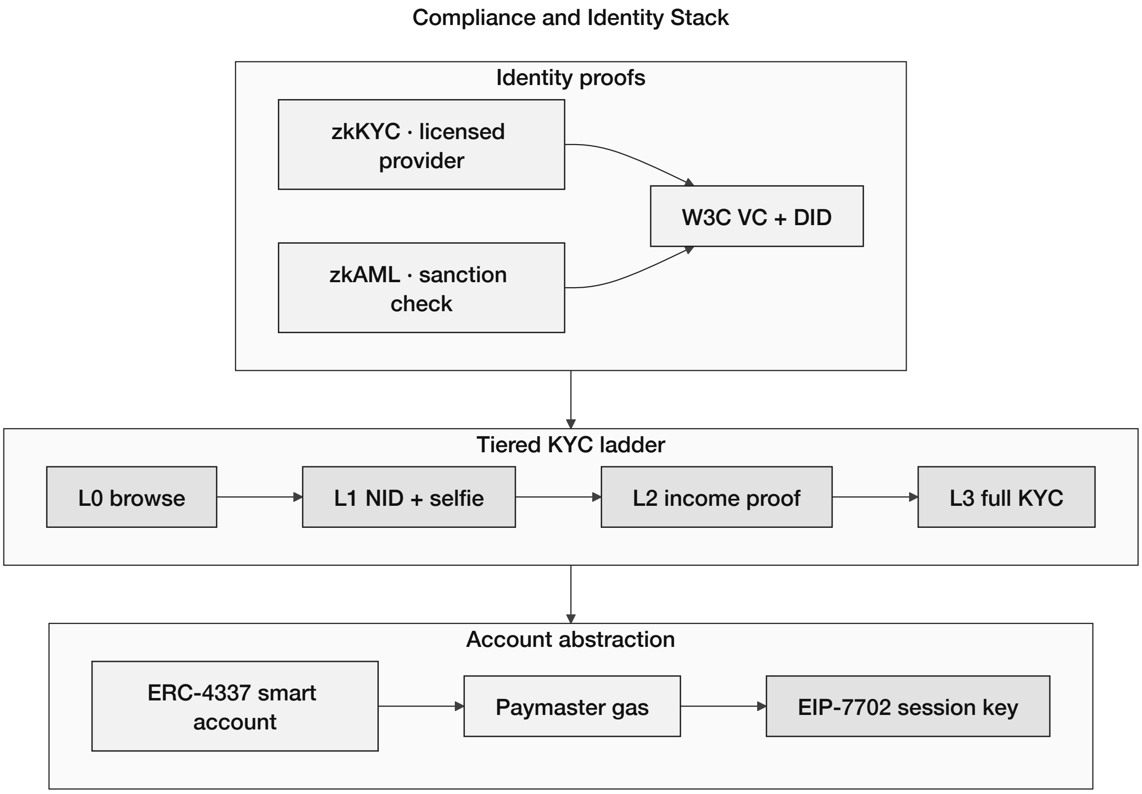


Figure 3.10: Compliance and identity stack

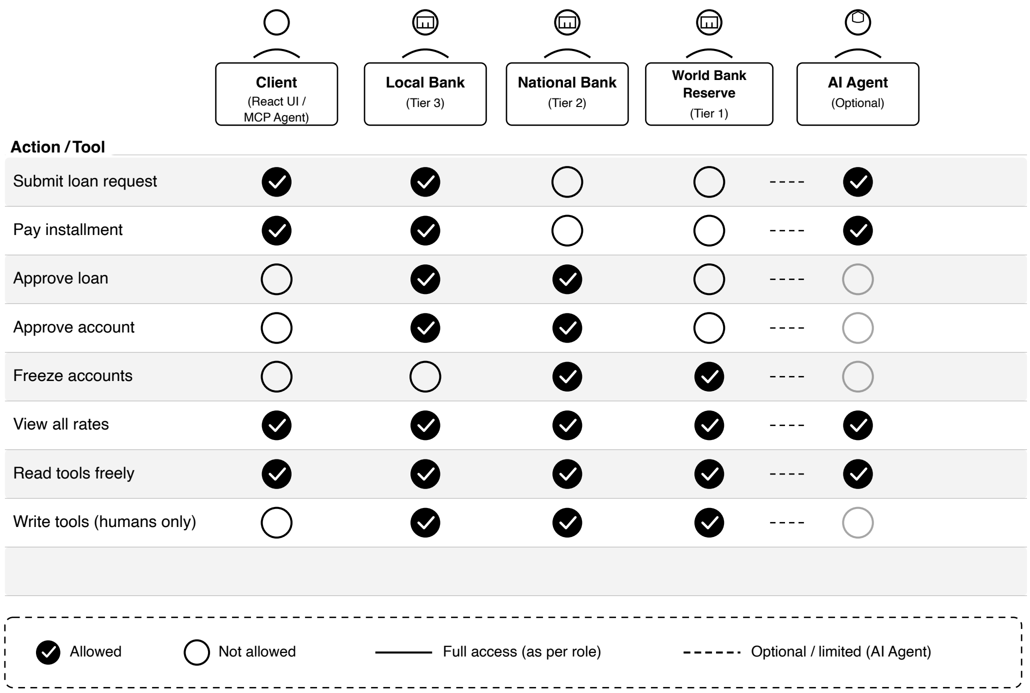


Figure 3.11: Actor permission matrix (primary actions)

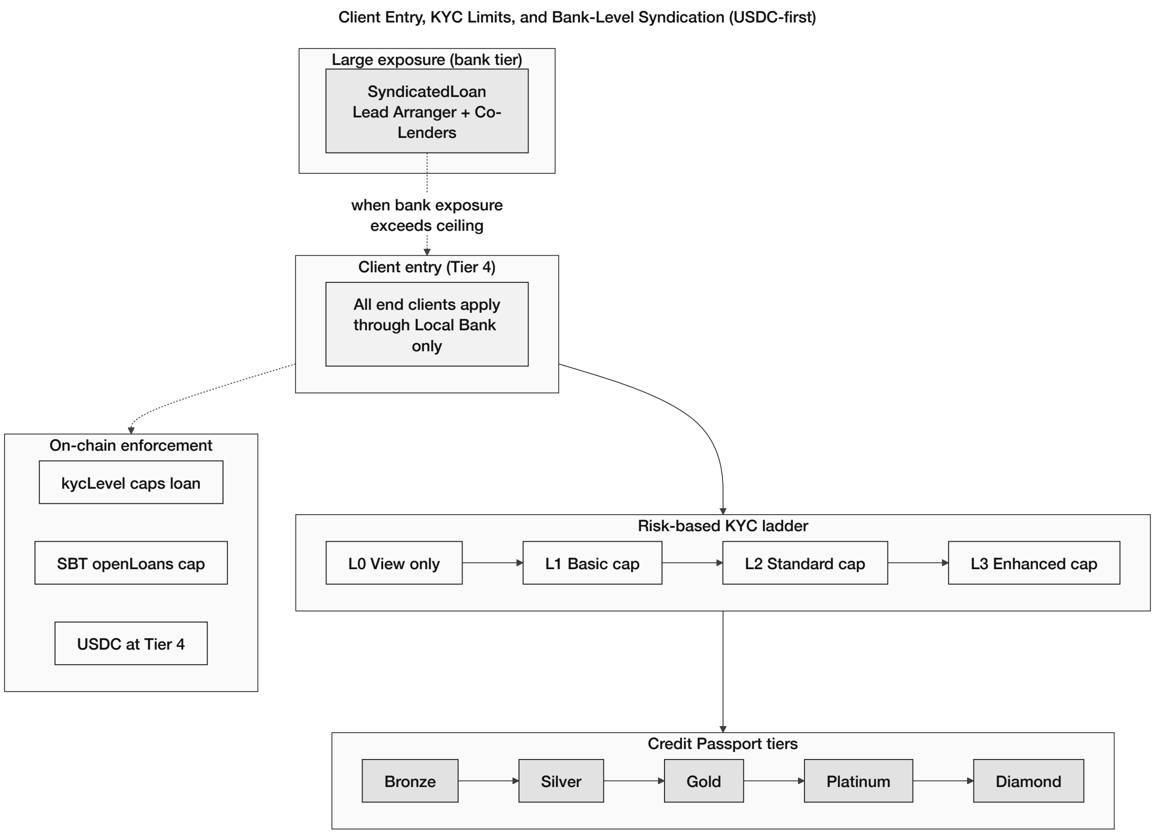


Figure 3.12: Client limits and bank-level syndication

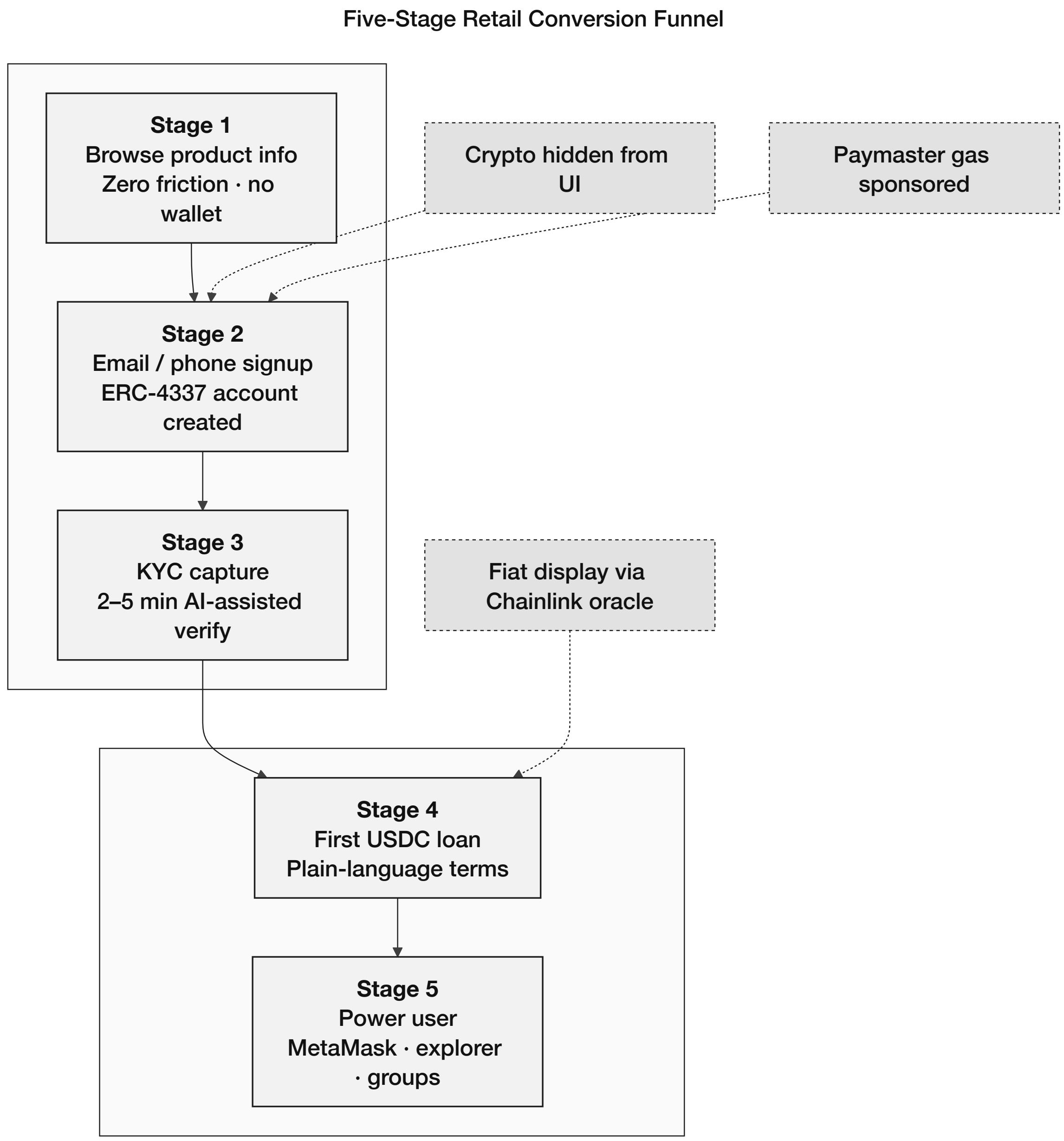


Figure 3.13: Five-stage retail onboarding funnel

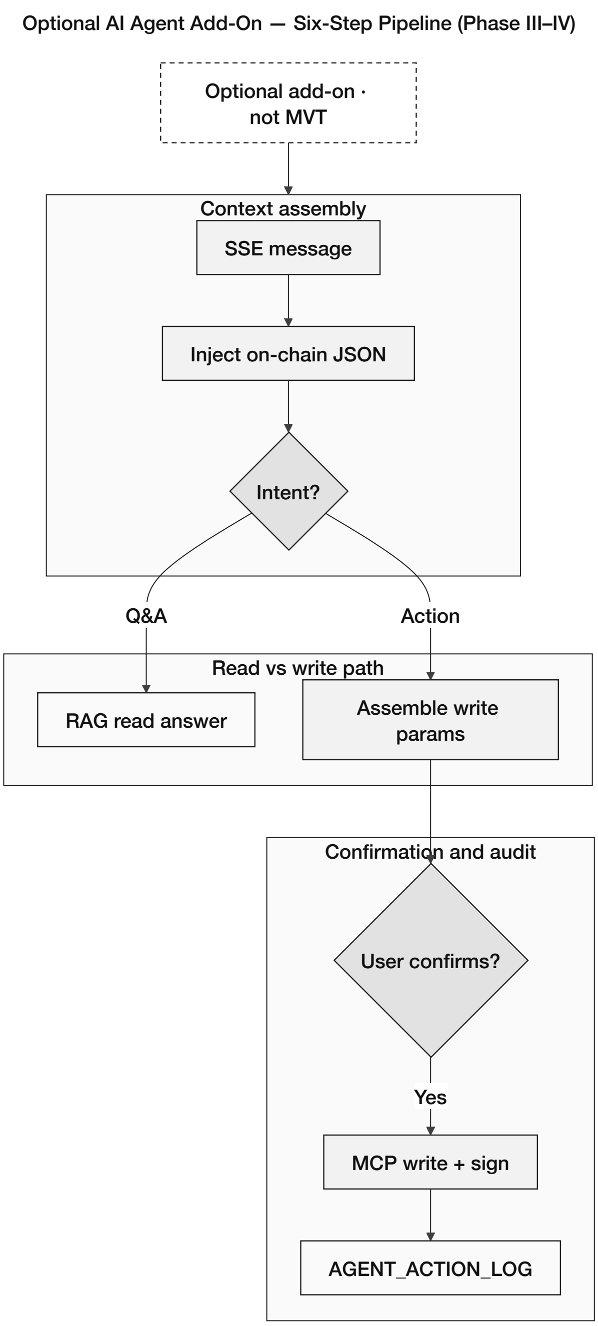


Figure 3.14: Optional AI agent add-on (Phase III–IV, excluded from MVT)

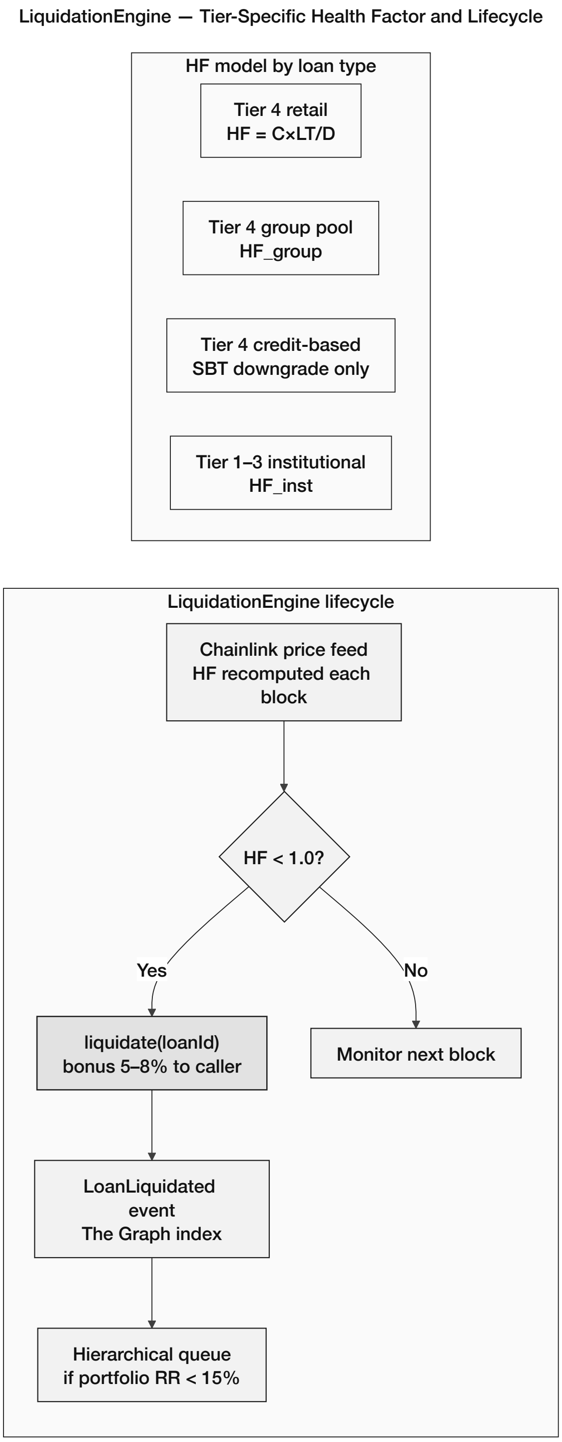


Figure 3.15: LiquidationEngine tier models and lifecycle

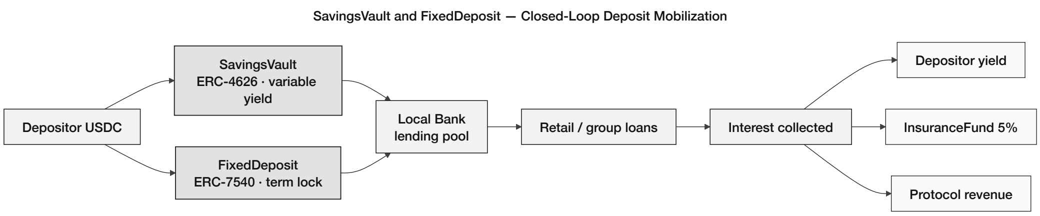


Figure 3.16: SavingsVault and FixedDeposit closed loop

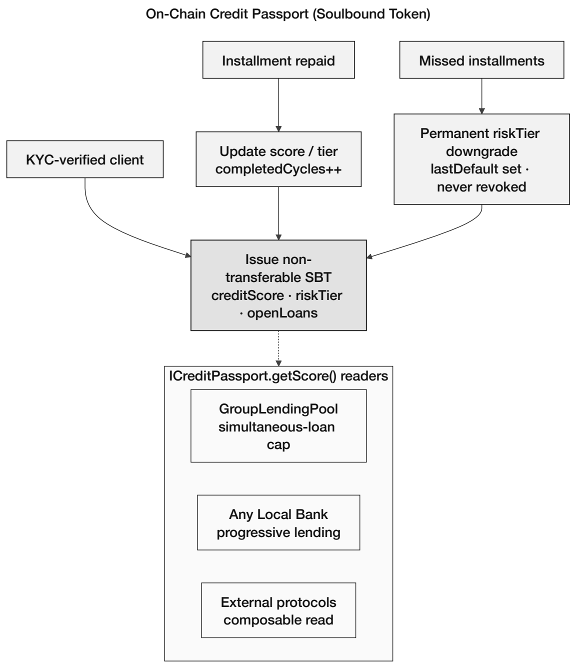


Figure 3.17: On-chain Credit Passport SBT lifecycle

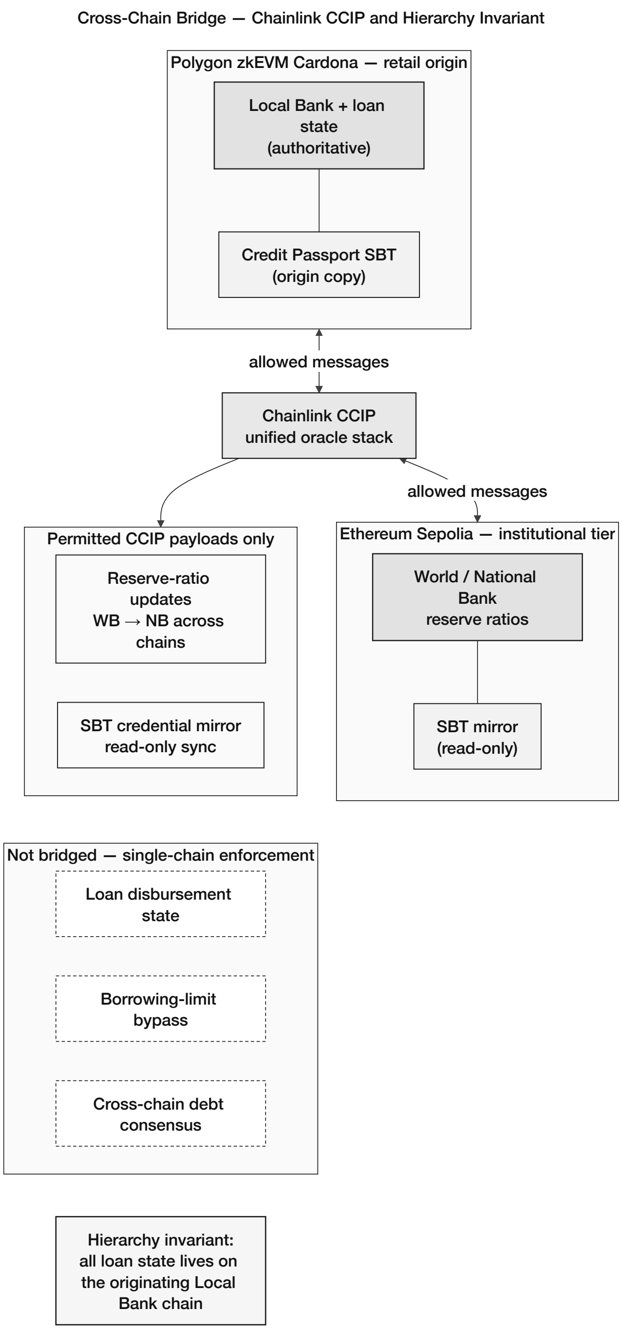


Figure 3.18: Chainlink CCIP cross-chain hierarchy invariant

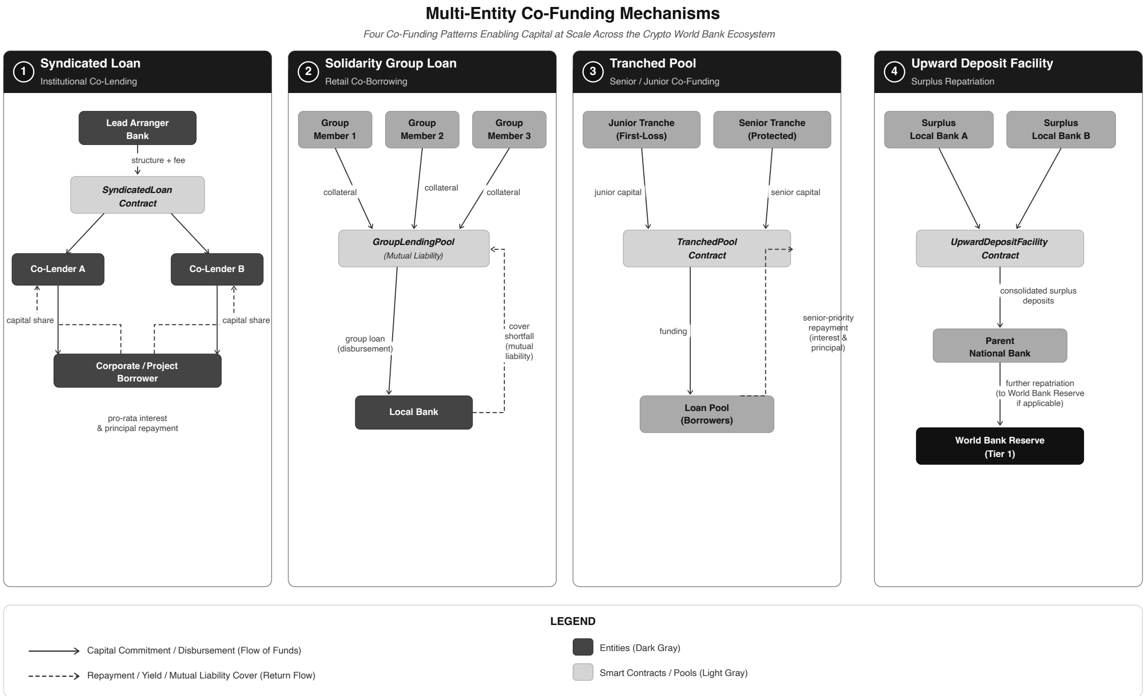
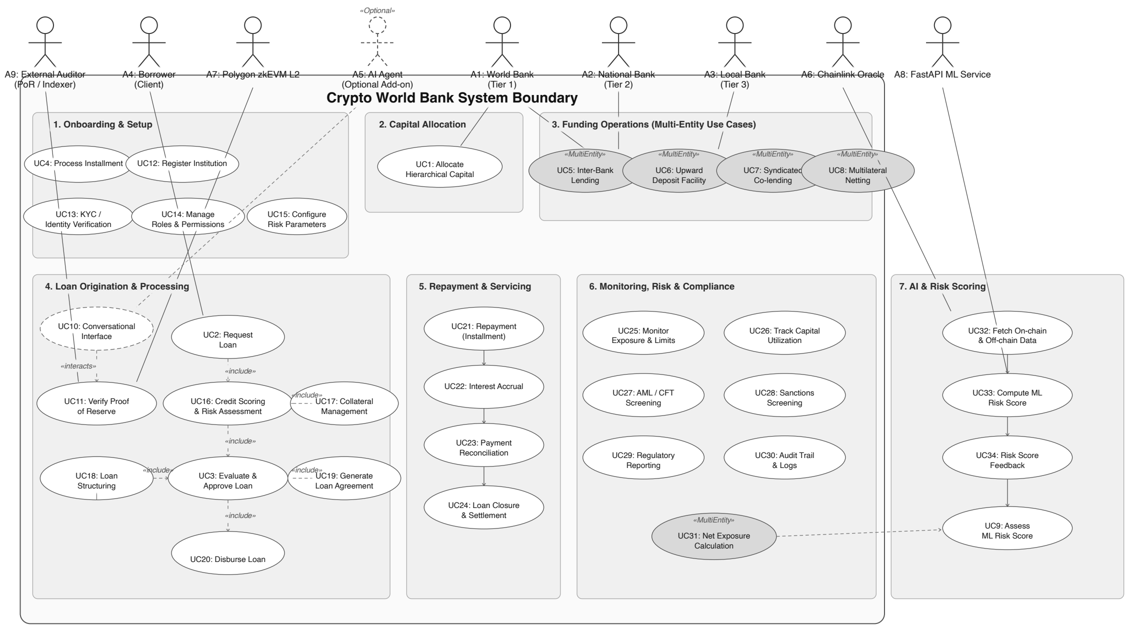


Figure 3.19: Multi-entity and cross-tier capital operations



Status		Definition	Applies To	Notes
Specified	(Formal Design)	Requirements and design specified in this document. Basis for evaluation.	All Use Cases & Actors defined above	Not implemented yet.
Planned	(Implementation)	Planned for implementation in later phases of the final thesis.	All Use Cases & Actors defined above	Implementation pending thesis phases.
Optional	(Add-on)	Optional conversational agent interface (UC10 and A5 only). Not part of core.	UC10 and A5 (AI Agent) only	May be implemented depending on scope and feasibility.

Note: Nothing in the pre-thesis submission is deployed to a public network; this diagram records design status only.

Figure 3.20: Use-case diagram (nine-actor taxonomy)

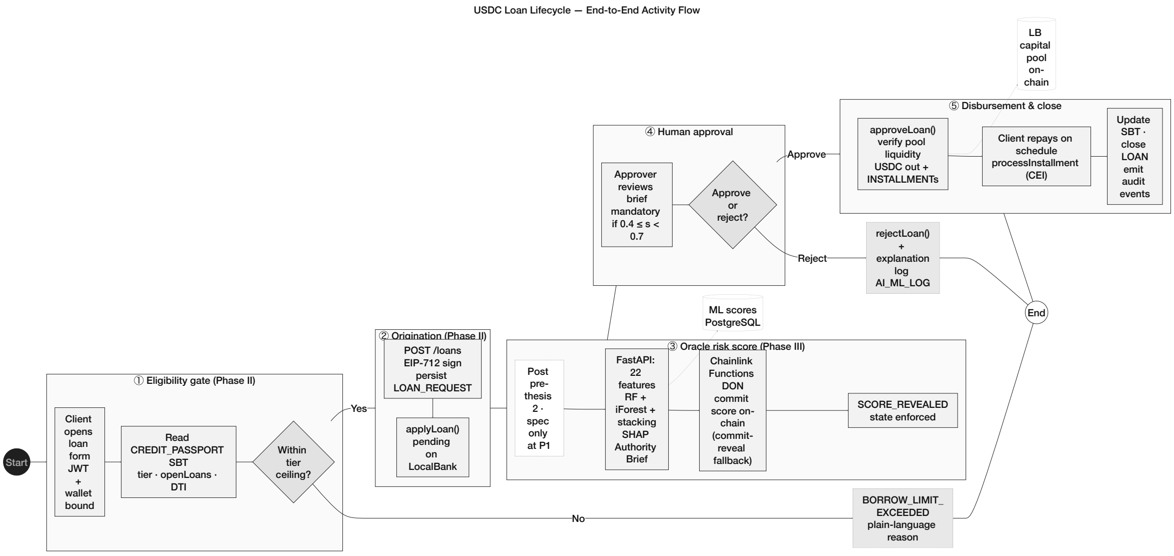


Figure 3.21: USDC loan lifecycle activity flow

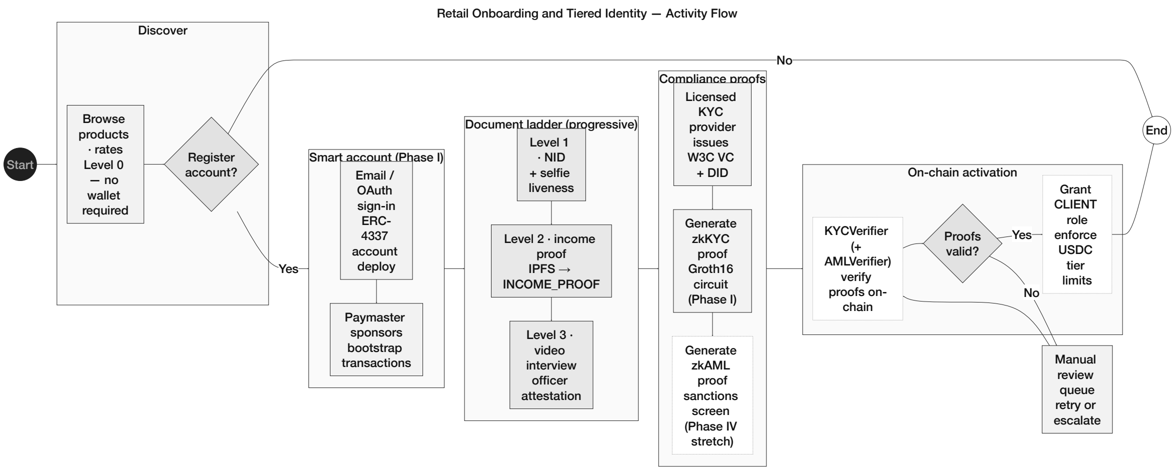


Figure 3.22: Retail onboarding and identity activity flow

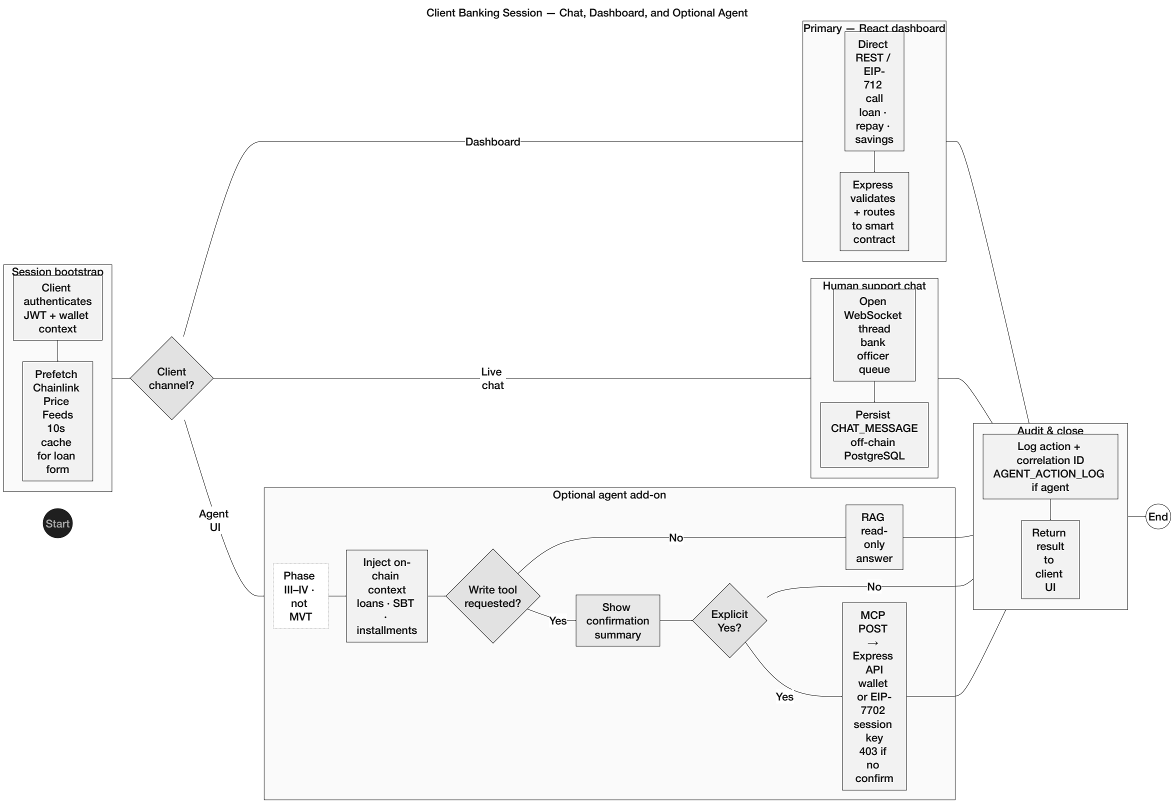


Figure 3.23: Client banking session activity flow

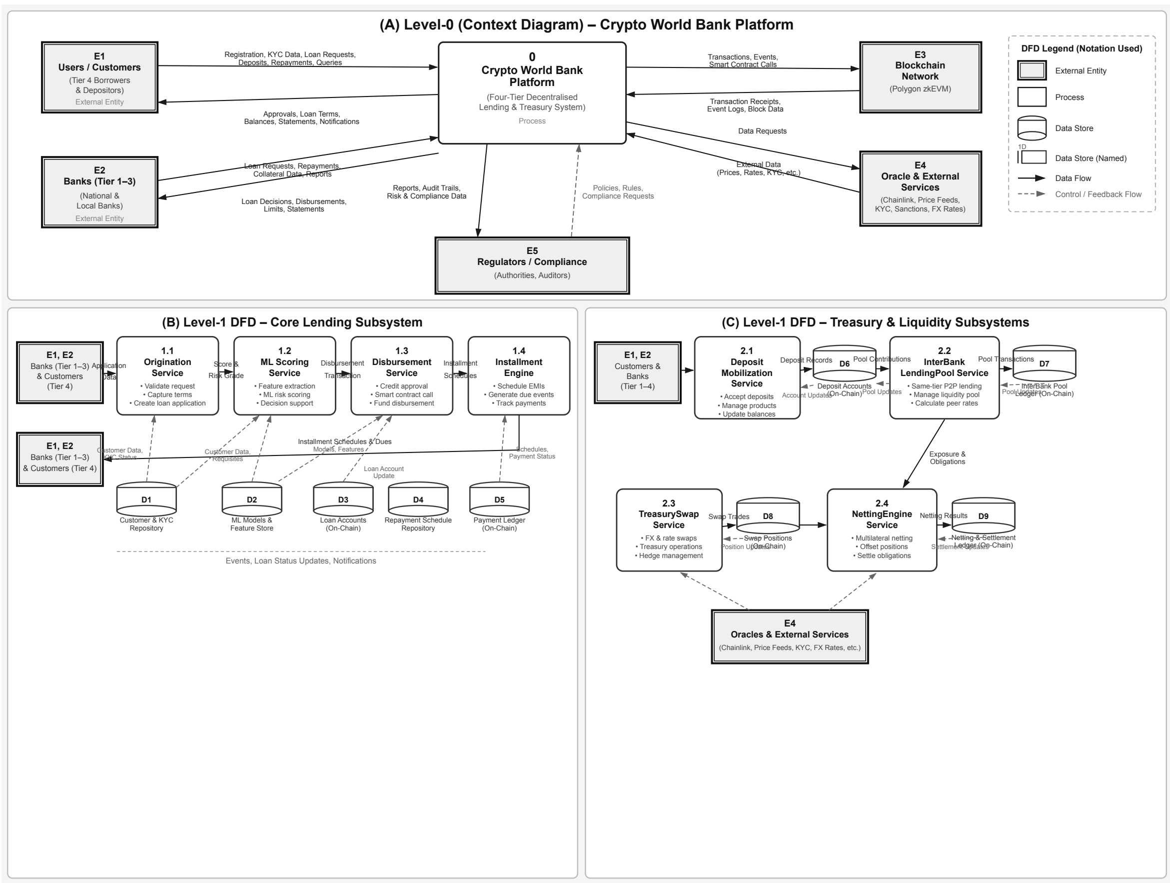


Figure 3.24: Data-flow diagrams

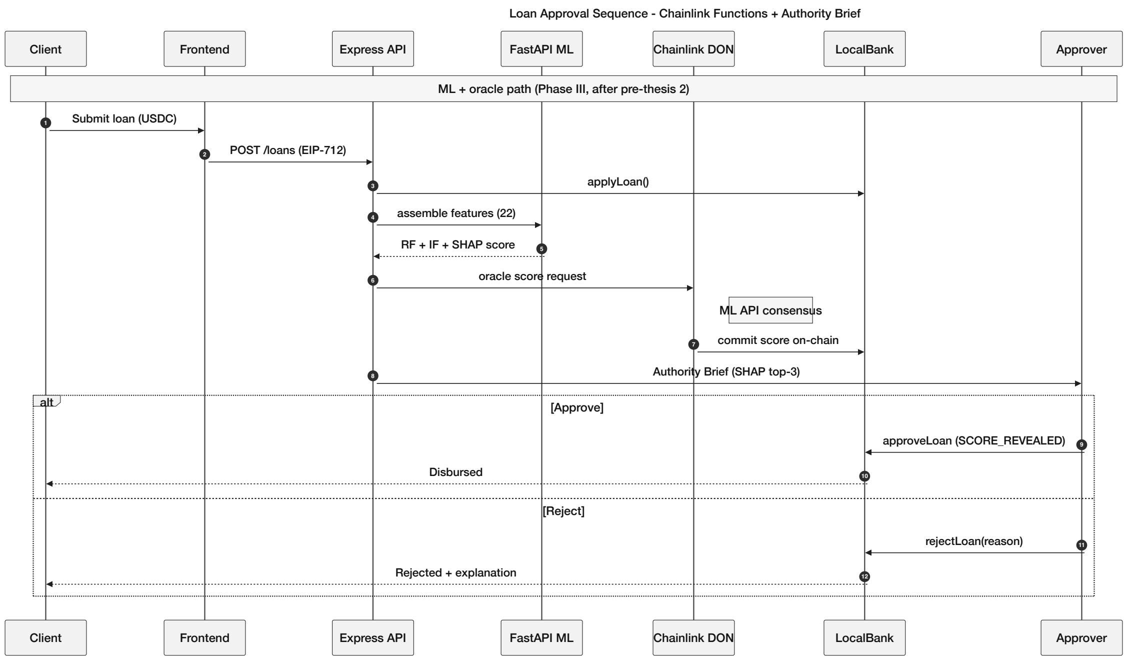


Figure 3.25: Loan approval sequence diagram

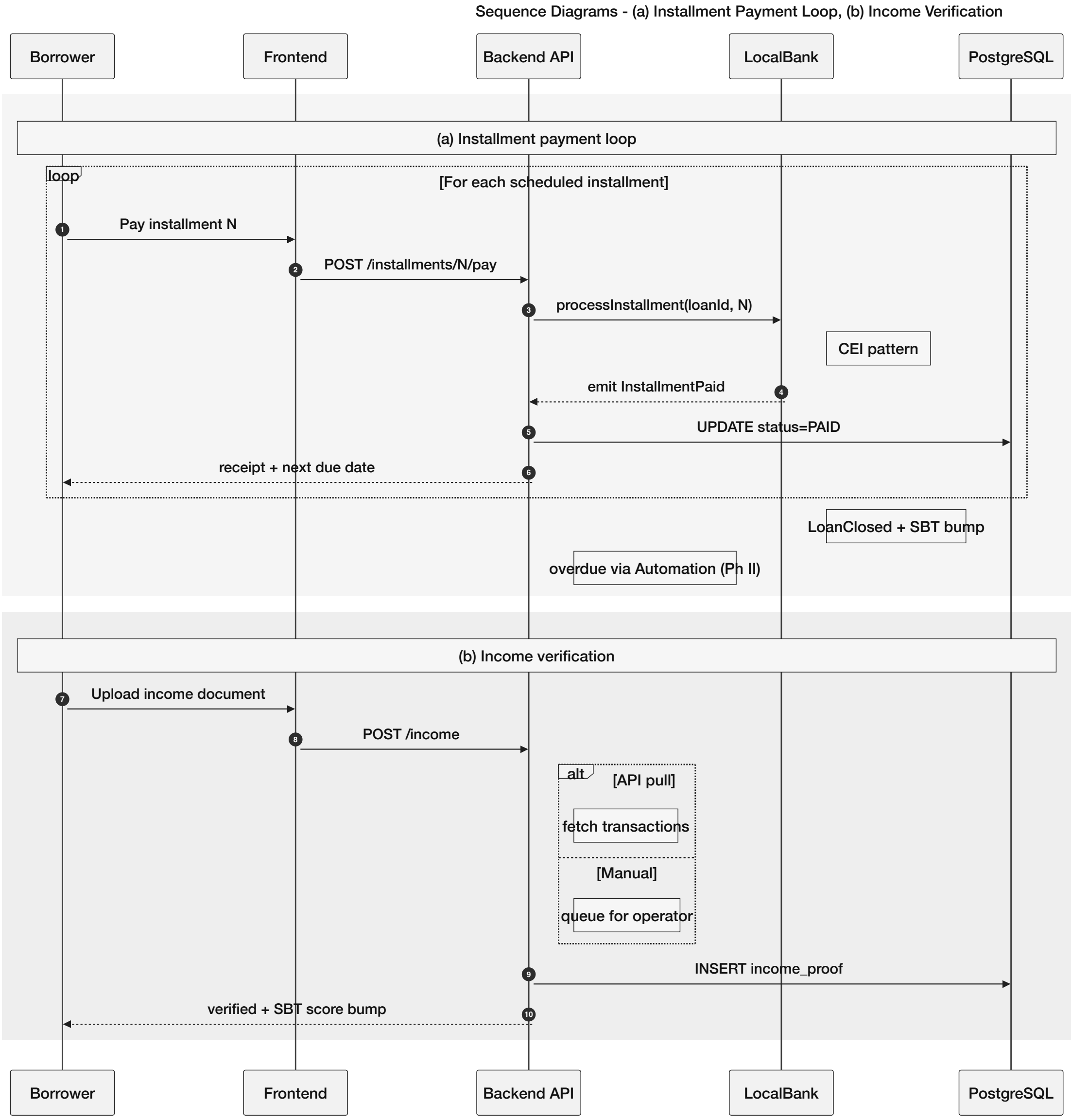


Figure 3.26: Sequence diagrams

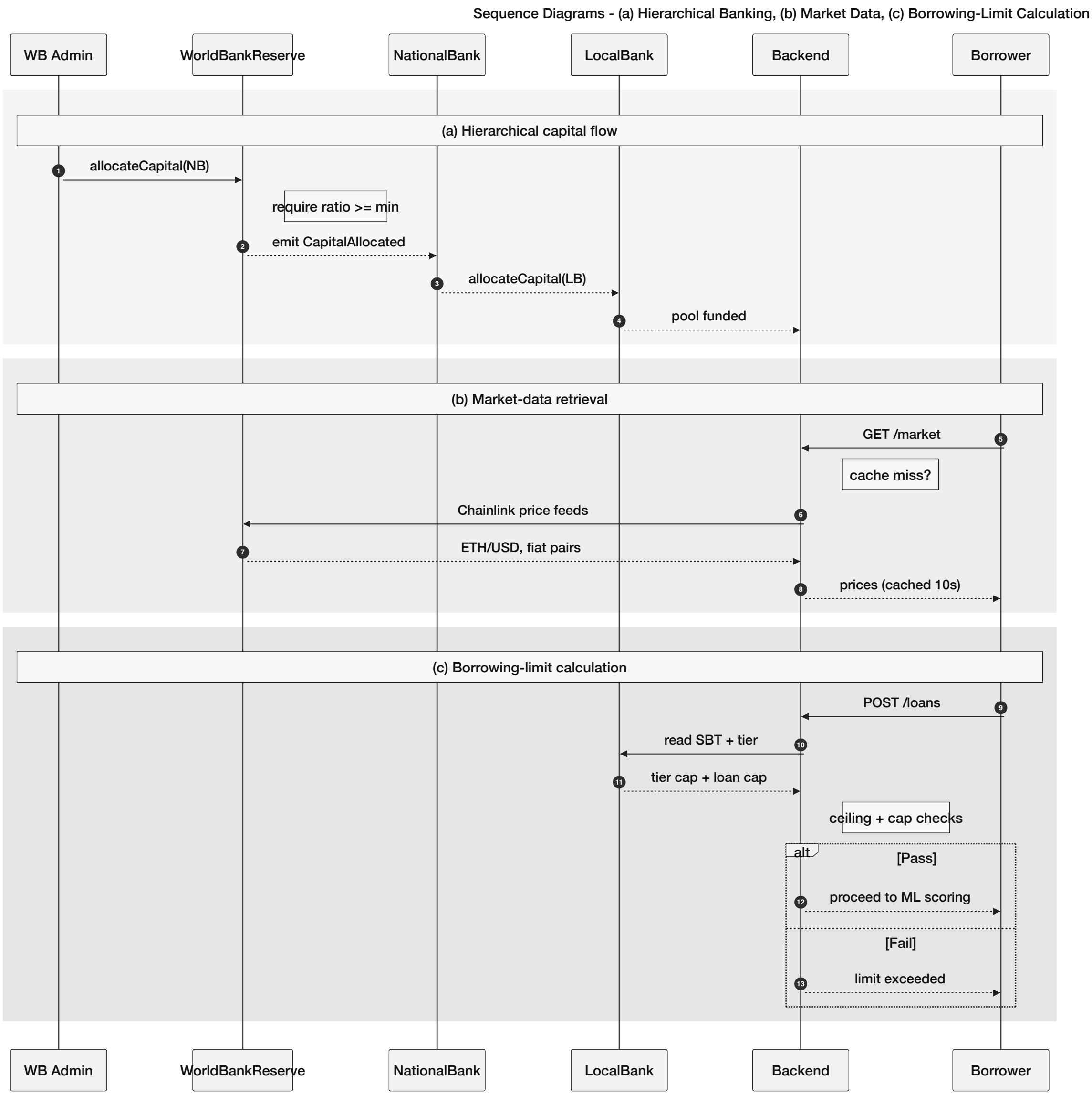


Figure 3.27: Sequence diagrams

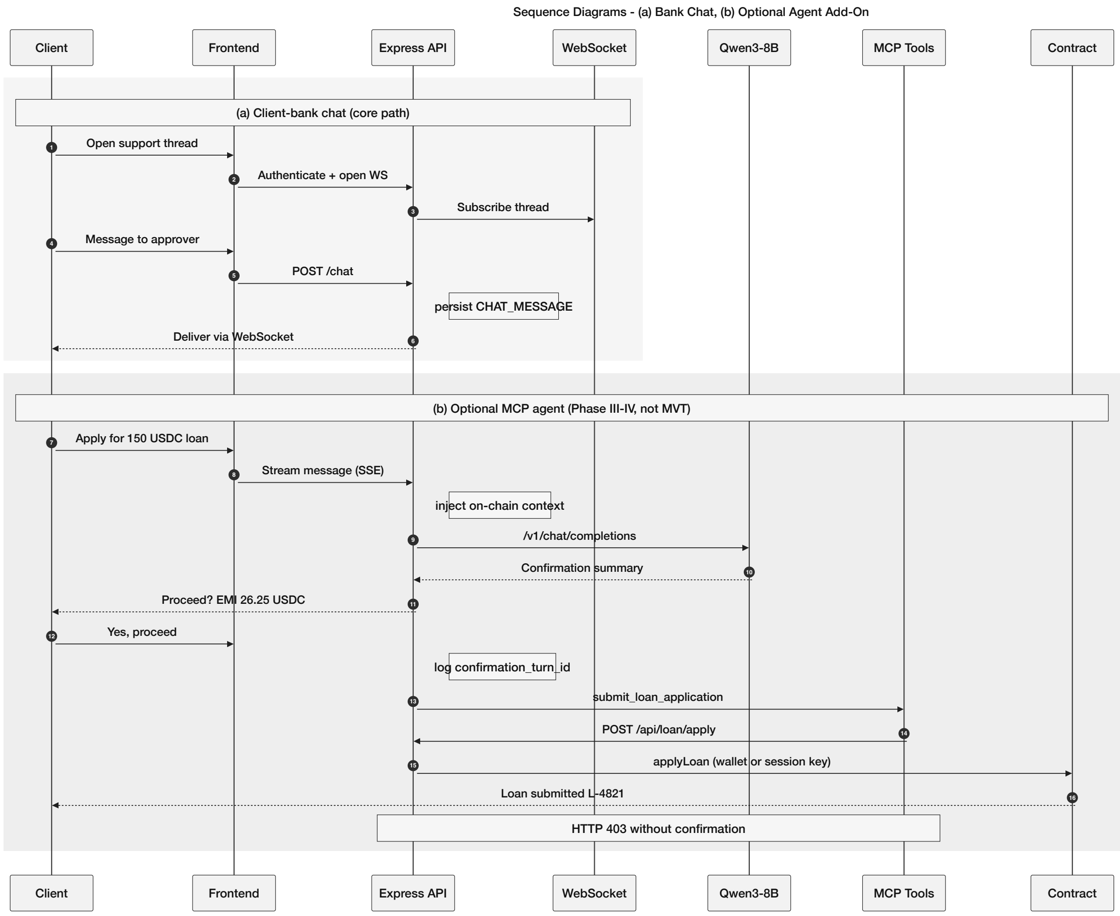


Figure 3.28: Sequence diagrams

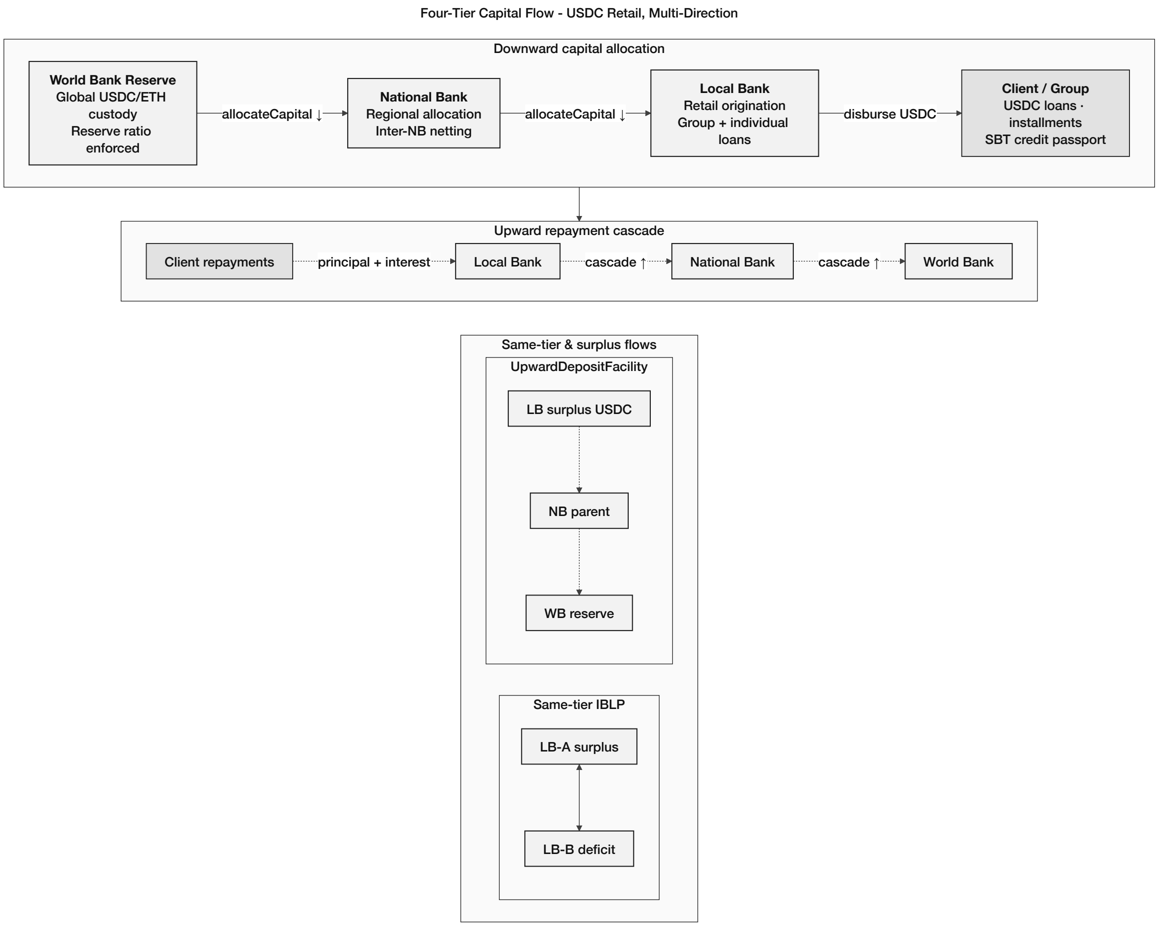


Figure 3.29: Four-tier hierarchical capital flow

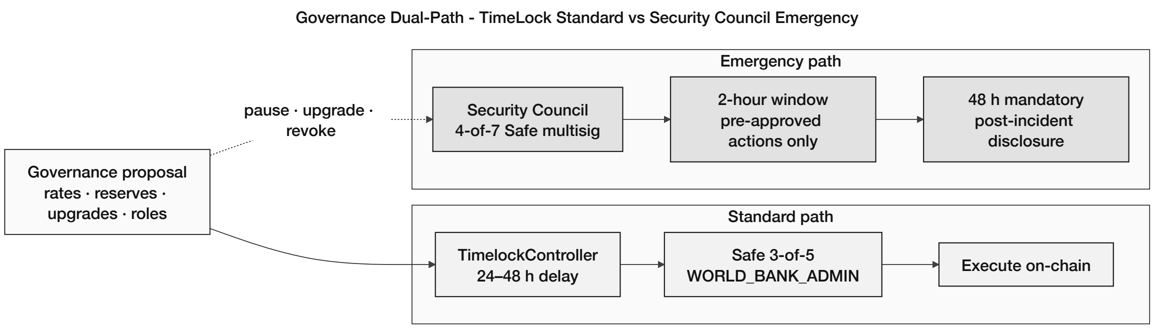


Figure 3.30: Governance dual-path

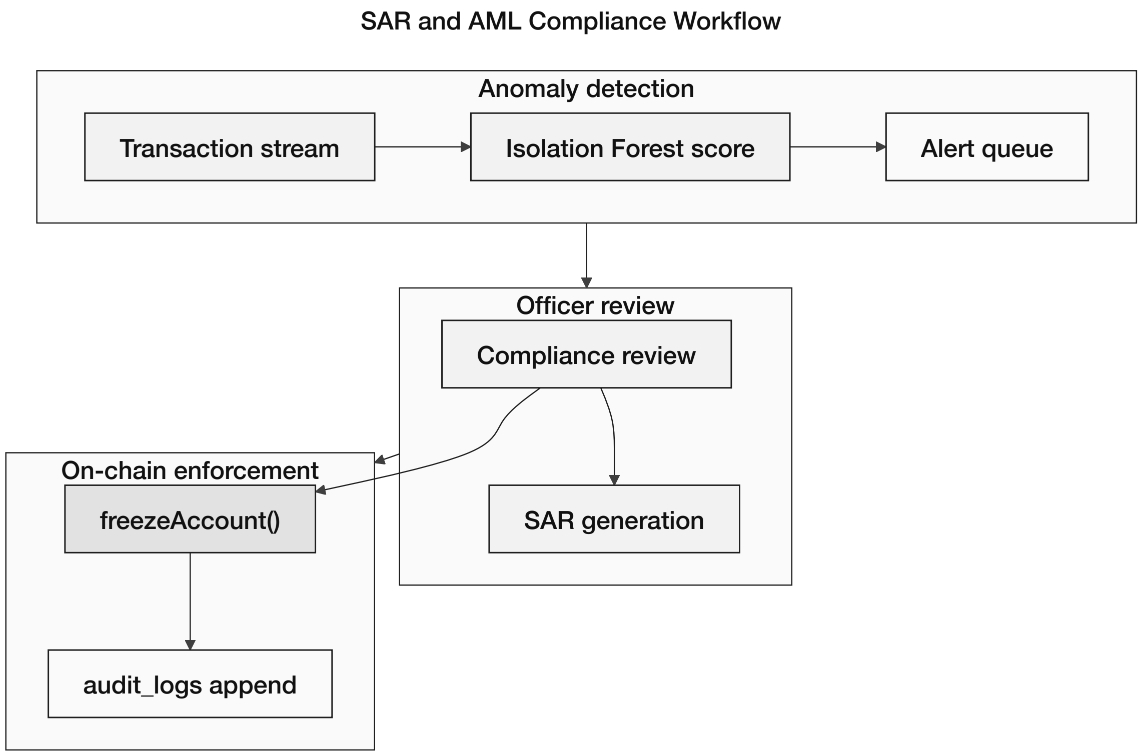


Figure 3.31: SAR and AML compliance workflow

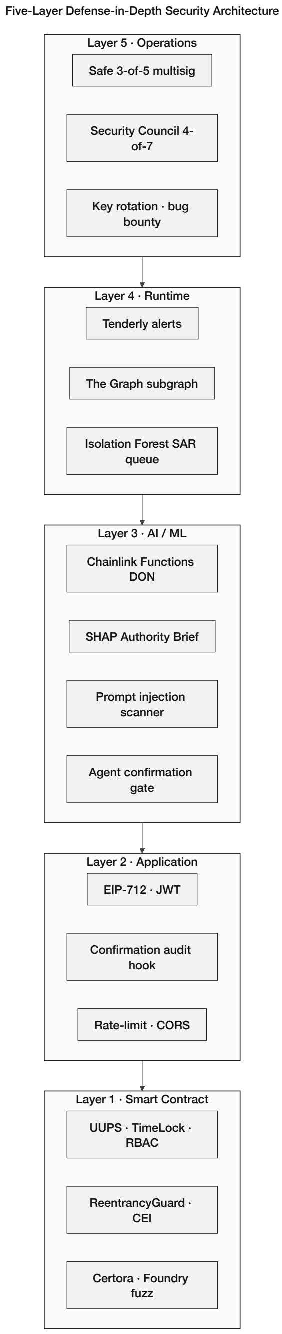


Figure 3.32: Five-layer defense-in-depth security architecture

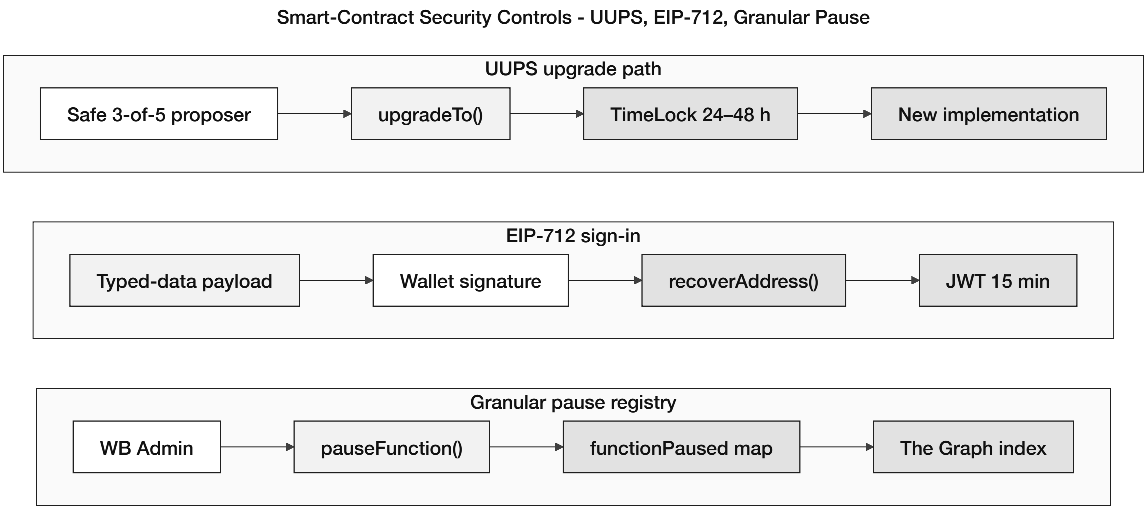


Figure 3.33: Smart-contract security controls

Chapter 4

Methodology

Lightweight **Agile/Scrum** across four implementation phases. **Core thesis** = on-chain banking + oracle ML; conversational agent is optional (Could-have).

4.1 MVT Checklist and Phase Plan

Graded pass line: (C1) four-tier contracts on testnet; (C2) `GroupLendingPool` spec + PoC; (C3) RF/IF/SHAP + Authority Brief after pre-thesis 2. Pre-thesis 1 delivers specification only—no trained models or live deploy.

Table 4.1: Four-phase implementation summary

Phase	Focus	Weeks	Key deliverables
I	Foundation	1–4	Contract hierarchy, DB schema, React shell
II	Core banking	5–9	Loan E2E, savings, group pool, automation
III	ML + oracle	10–13	Chainlink Functions, Authority Brief, dashboard
IV	Verification	14–16	Foundry sim, Certora stretch, testnet manifest

Table 4.2: Core vs. optional capabilities

Capability	Priority	Phase / interface
Four-tier smart contracts	Must (I–II)	React + wallet; Foundry tests
On-chain loan lifecycle	Must (II)	`LoanController` state machine
Chainlink Functions oracle	Must (III)	FastAPI → DON → on-chain score
RF + IF + SHAP	Must (III)	Authority Brief; audit log
MCP conversational agent	Could (III–IV)	Alternate API path only

4.2 Design Decisions

Table 4.3: Design decisions (selected)

Area	Choice	Alternative	Criterion
Network	Polygon zkEVM Cardona	Amoy PoS	ZK validity proofs, low gas
Contracts	Solidity EVM	Solana	Tooling, testnet cost
Fraud ML	Random Forest + IF	XGBoost	SHAP compatibility
Oracle	Chainlink Functions	Commit-reveal only	Decentralised trust
Database	PostgreSQL	SQLite	3NF, async API

4.3 Methodology Diagrams (Full Set)

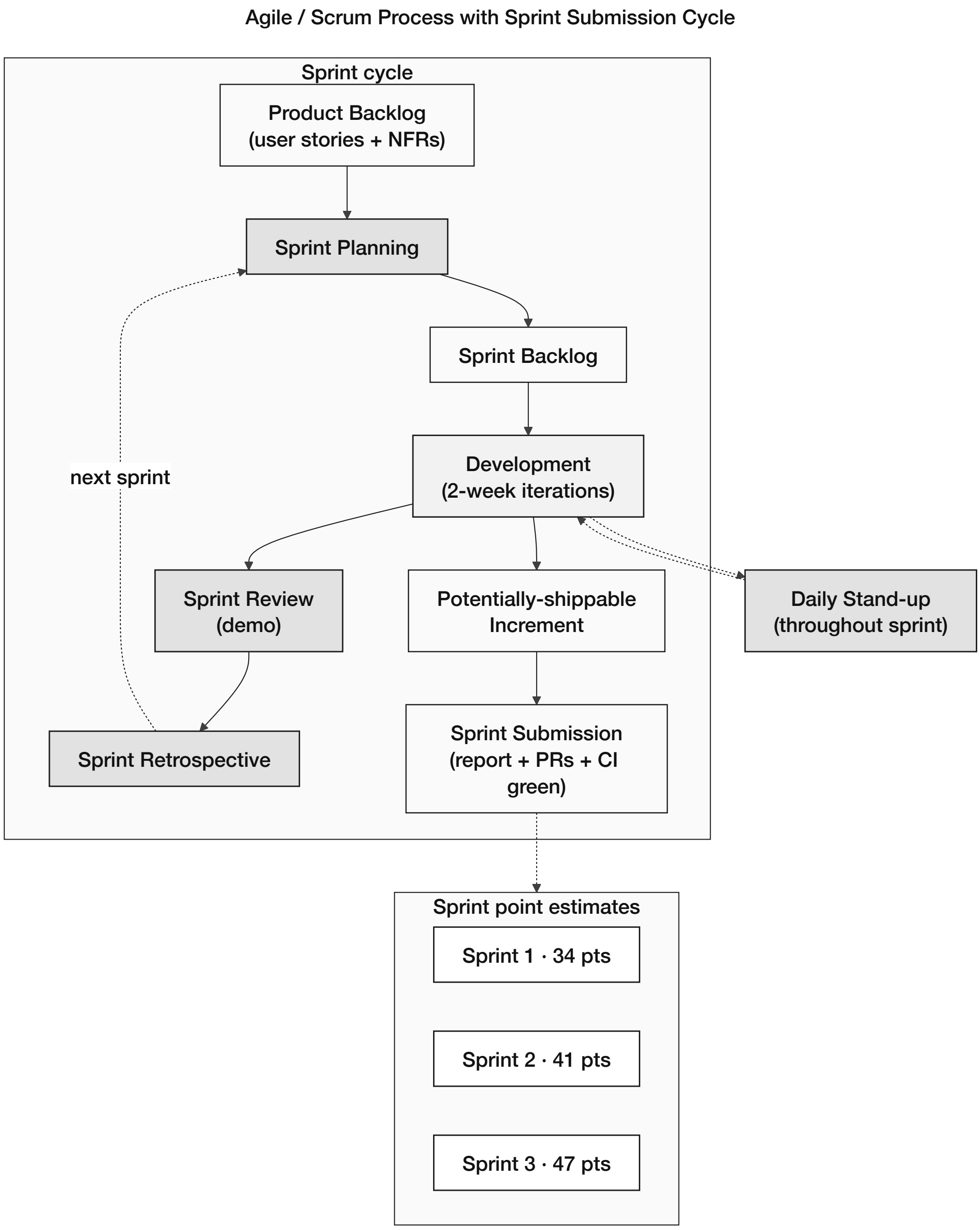


Figure 4.1: Agile / Scrum development process

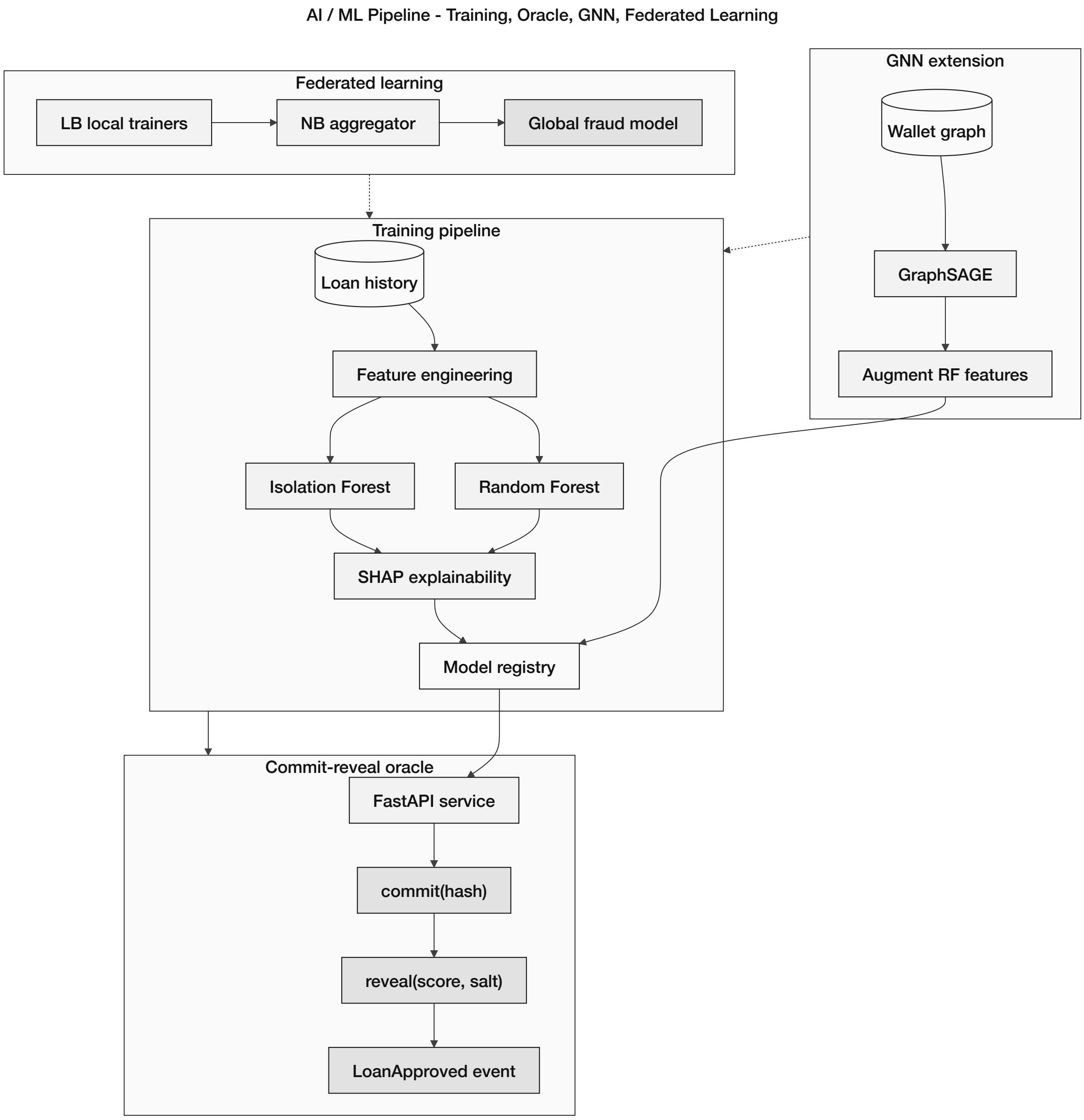


Figure 4.2: AI/ML pipeline wiring

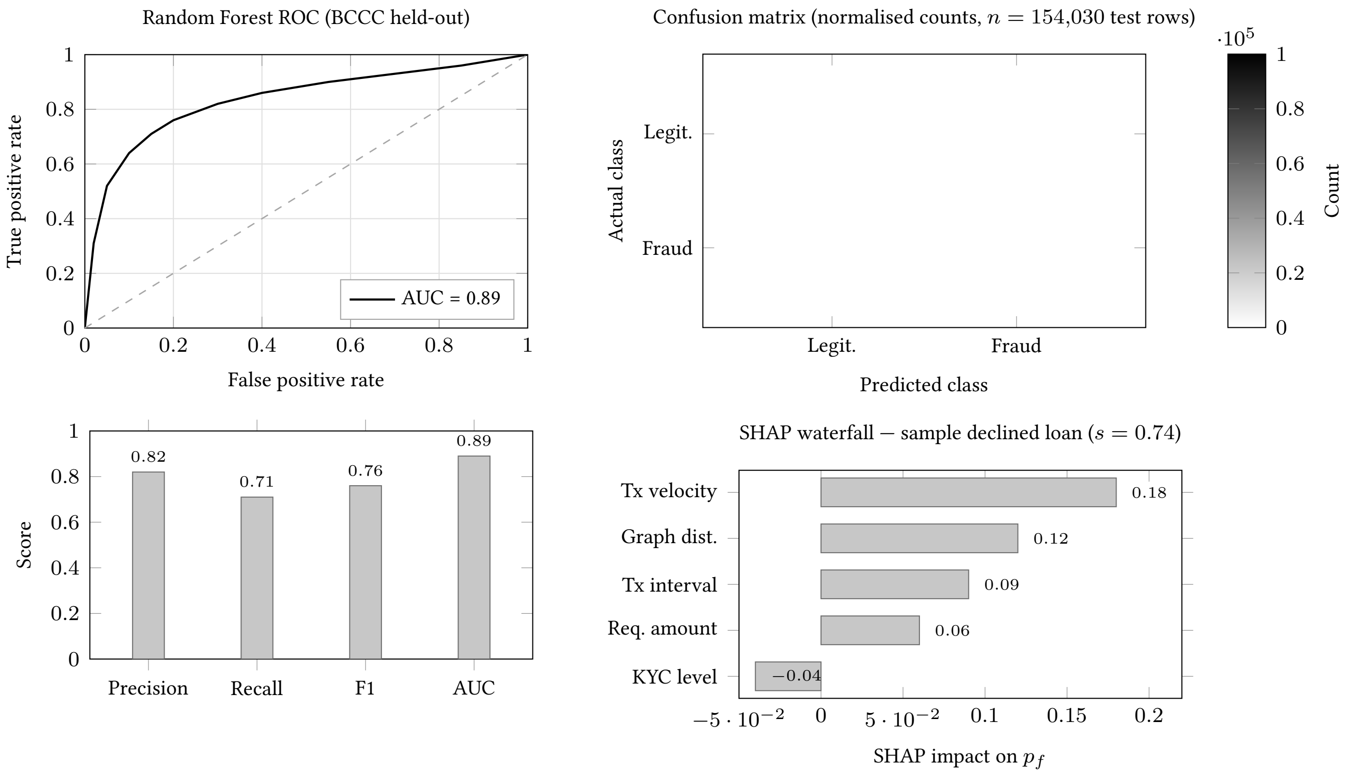


Figure 4.3: Planned ML evaluation layout (post pre-thesis 2)

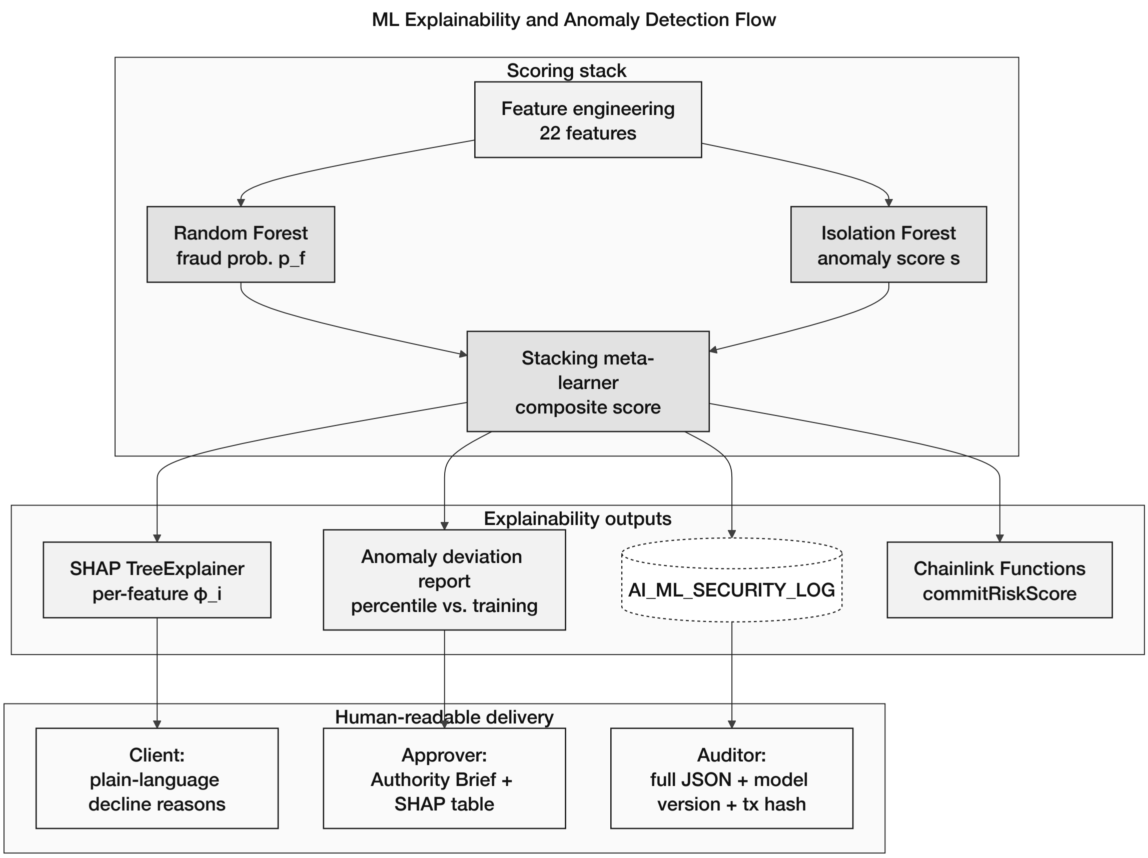


Figure 4.4: Explainability and anomaly-detection flow

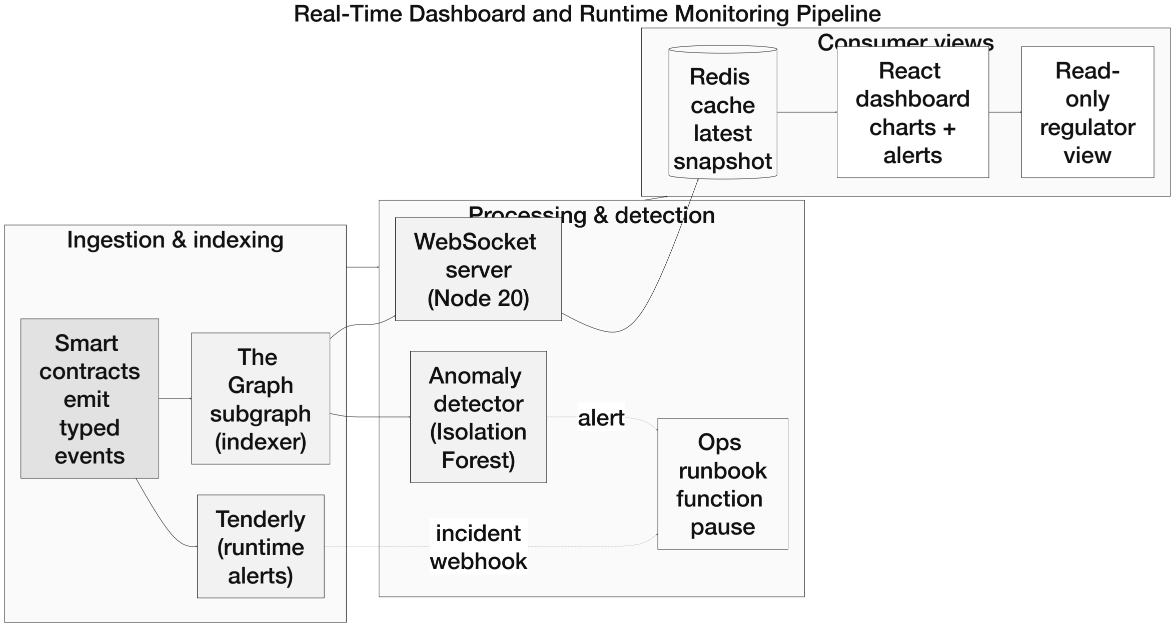


Figure 4.5: Real-time dashboard and runtime monitoring pipeline

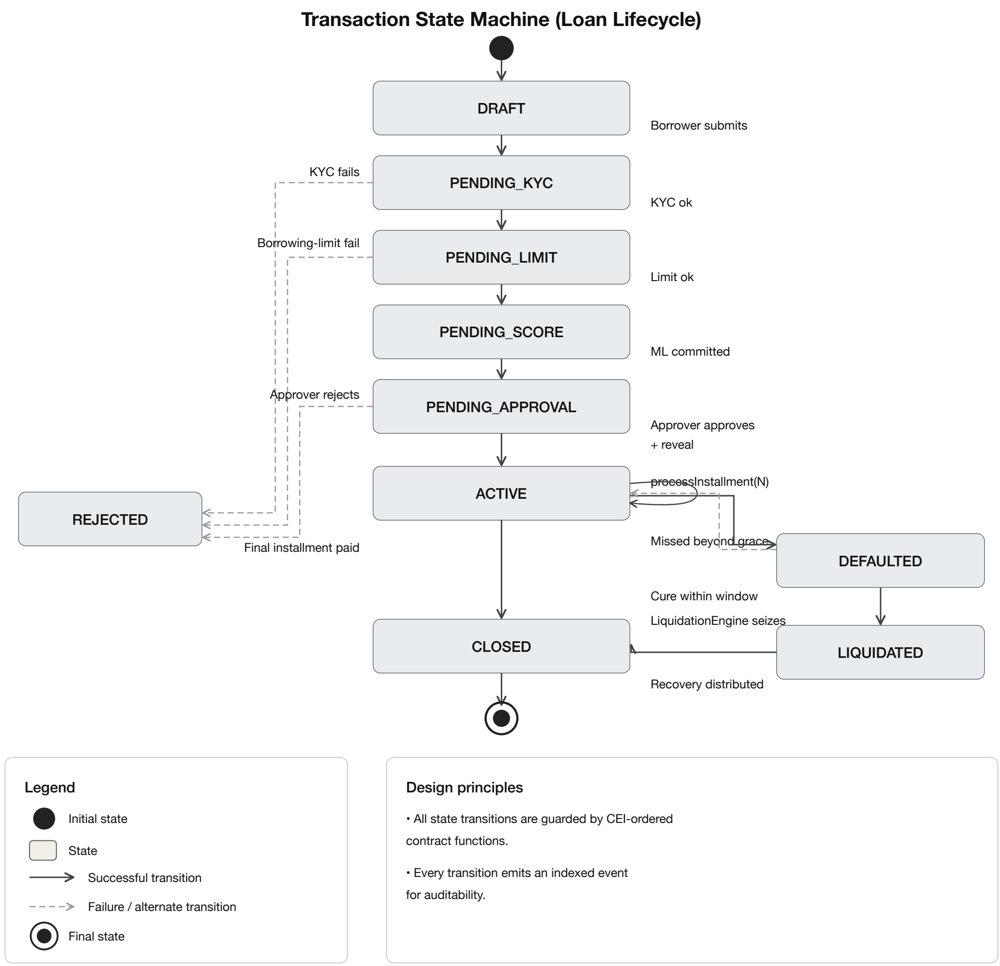


Figure 4.6: Transaction state machine of the loan lifecycle

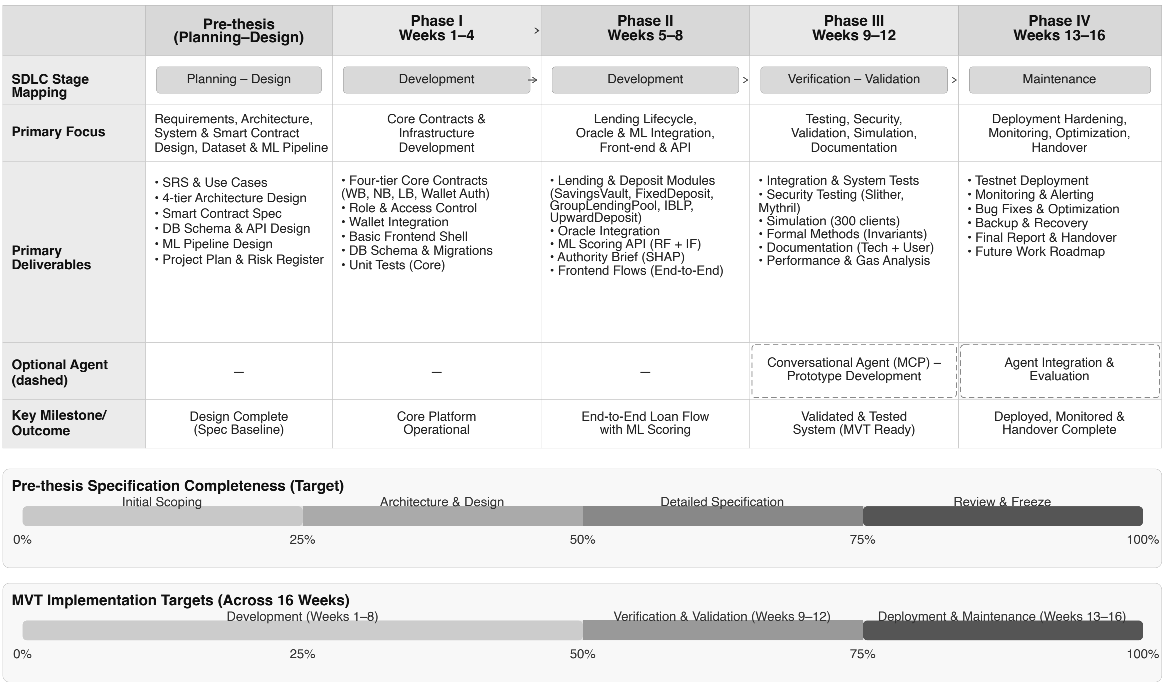


Figure 4.7: Four-phase roadmap with SDLC mapping and progress

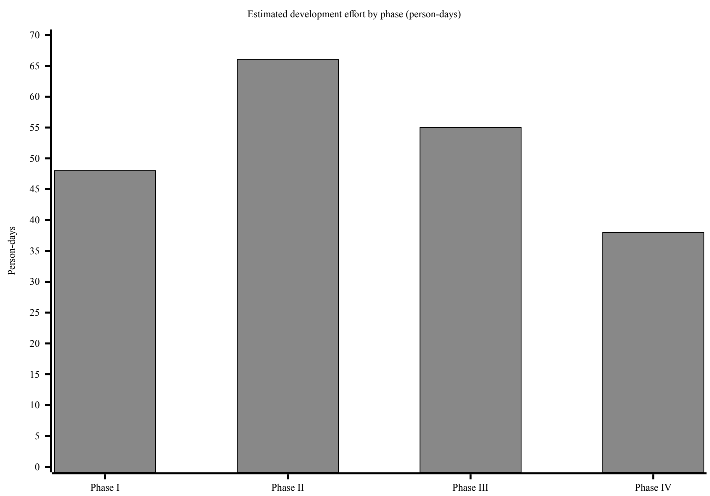


Figure 4.8: Development effort by implementation phase

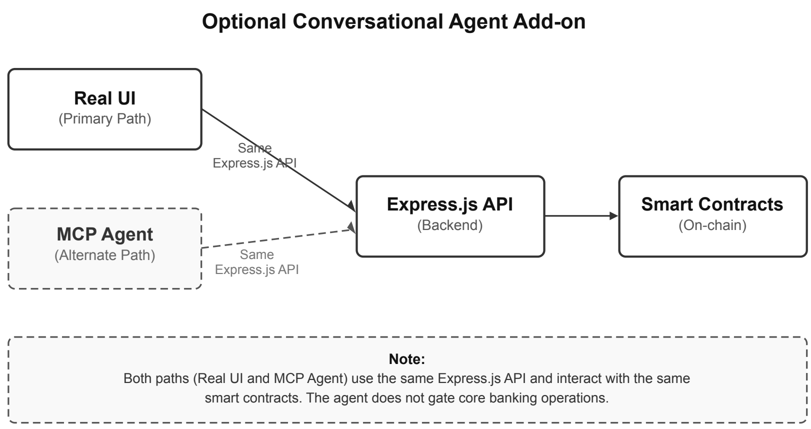


Figure 4.9: Optional conversational agent add-on

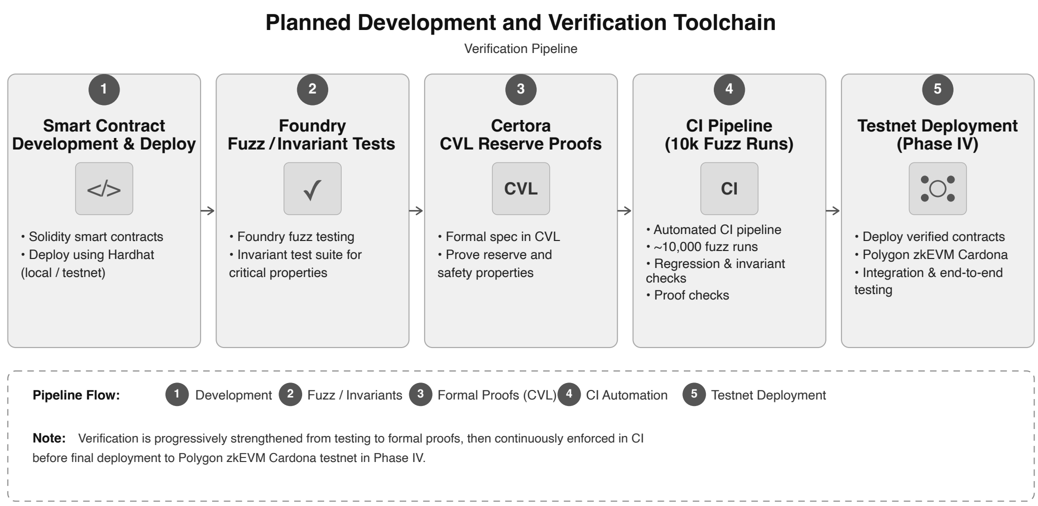


Figure 4.10: Planned development and verification toolchain

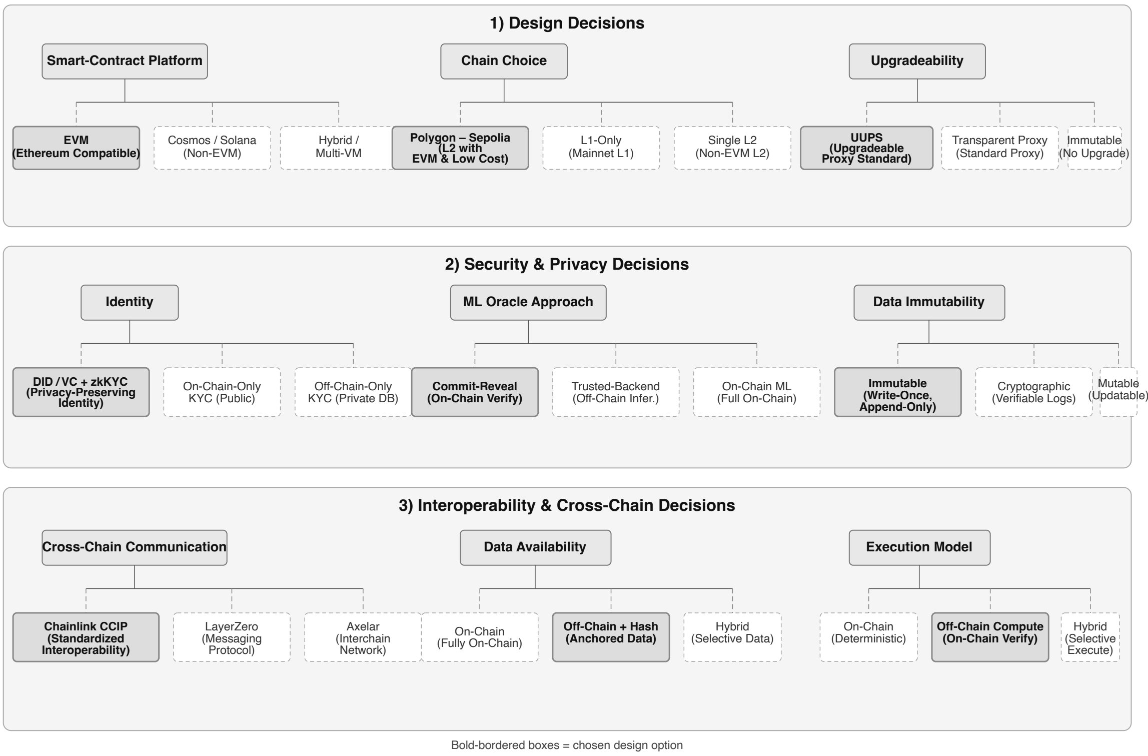


Figure 4.11: Key design decisions and alternatives considered

Chapter 5

Evaluation and Discussion

5.1 Performance Evaluation

Our pre-thesis 1 shows specifications and plans and creates checkboxes, not any kind of live benchmarks, since the project has not been deployed or tested yet. Project influential elements and financial feasibility are further elaborated in Section 5.4. Our planned RQ-level evaluation will be executed after pre-thesis 2. Our plan is for projected gas per loan to stay below interest achieved on L2; the result of default stress is given in Table 5.7.

5.2 Analysis of Design Solutions

Our choices of technology stack were chosen based on ecosystem maturity, as per our goal to create a system with zero-cost prototype constraints and more transparent auditability (visible in Table 4.3, Section 4.2). The chosen stack for off-chain pairs EVM Solidity on PostgreSQL + FastAPI, zkEVM for L2 execution, and to connect to on-chain we use Chainlink Functions attached with ML scores. The optional conversational MCP agent and EIP-7702 session keys remain Phase III–IV extensions outside the MVT.

5.3 Final Design Adjustment

The final pre-thesis 1 outlook locks in four adjustments before implementation. These are stablecoins, which is the first retail lending path and is used to avoid ETH liability volatility (USDC/USDT, Section 1.7). After that, L2 deployment for micro-loan gas economics. The third one is ML as decision support only, meaning human approvers and on-chain gates, no auto-disbursement. Last but not least is MVT scope. It is limited to the three-tier lending core, database schema, and dashboard; syndicated modules, full regulatory filing, and the conversational agent are deferred.

5.4 Feasibility Analysis

5.4.1 Technical Feasibility

Each layer is specified against mature tooling; nothing requires bespoke infrastructure beyond standard testnets and open-source stacks (Table 5.1).

Table 5.1: Technical feasibility assessment

Component	Assessment	Evidence
Smart contracts	Feasible (specified)	Solidity; Hardhat; OpenZeppelin; Phases I–II
Frontend DApp	Feasible (specified)	React 18 + Wagmi/RainbowKit
Blockchain deployment	Feasible (planned)	Polygon zkEVM Cardona + Sepolia
AI/ML integration	Feasible with constraints	RF + IF + SHAP; Chainlink Functions Phase III
Optional agent	Feasible (deferred)	MCP + Qwen3-8B; Phase III–IV; outside MVT
Database backend	Feasible	PostgreSQL 20-entity schema (3NF); FastAPI

5.4.2 Economic Feasibility

Table 5.2: Economic feasibility — zero-cost prototype

Cost Category	Estimate	Notes
Blockchain deployment	$0	Public testnets
Frontend / backend hosting	$0	Vercel / Render free tiers or localhost
AI/ML training	$0	Local machine or Colab free tier
Development tools	$0	Hardhat, VS Code, Git (open-source)
Total prototype	**$0**	Pilot ops ≈$500–$2,000/month at scale

5.4.3 Risk Feasibility

Main delivery risks and mitigations are summarised in Table 5.3; smart-contract and regulatory items dominate severity.

Table 5.3: Risk taxonomy and mitigation

Risk Category	Description	Severity	Mitigation
Partner non-cooperation	Key partners decline to join	Medium	Testnet pilots; open-source replication
Smart contract vulnerability	Logic exploit	High	OpenZeppelin; audits; pause mechanism
Regulatory adversity	Jurisdiction blocks retail crypto	Medium	Testnet-only prototype; sandbox path
AI/ML model degradation	Fraud model drift	Low	Retraining; human-in-the-loop; SHAP review

5.4.4 Regulatory Feasibility

The CWB platform uses basic compliance measures for retail stablecoin lending like MiCA, GENIUS Act, and EU AI Act, using authorized stablecoin and keeping PII off-chain (Table 5.4). But for future work, full conformity files and national licensing are kept. This thesis only focuses on testnet.

Table 5.4: Regulatory design controls (selected)

Statute / Article	Requirement	CWB Control
MiCA Art. 16	Authorised EMT issuer	Uses USDC/USDT; does not issue
MiCA Art. 36	1:1 reserve backing	Issuer attestations before pool acceptance
EU AI Act (credit scoring)	High-risk AI from Aug. 2026	Human approver gate; SHAP + audit log
GDPR Art. 17	Right to erasure	PII off-chain only; hashes on-chain

5.5 Statistical Analysis

Based on planning assumption rather than platform data, the table below presents spread revenue, ETH-price sensitivity, and default stress at deployment scale. As shown in Figure 5.2, the interest structure follows the lending hierarchy of 3%, 5%, and 8%.

Table 5.5: Revenue projection assumptions

Parameter	Value
Reference ETH price	$2,500 (planning mid-point, Feb 2026)
World Bank → National Bank	3% APR
National Bank → Local Bank	5% APR
Local Bank → Client	8% APR
Average loan term	12 months
Default rate provision (base)	8%
Origination fee	0.25% per disbursement

Table 5.6: ETH-price sensitivity of spread revenue

ETH Price	Loan Base	Spread Revenue	Incl. Fees (est.)
$2,000	$1.376B	$110.1M	$120–128M
$2,500 (base)	$1.720B	$137.6M	$150–159M
$3,000	$2.064B	$165.1M	$180–191M

Table 5.7: Default-rate sensitivity

Scenario	Default rate	Annual loss	Basis
Optimistic	3.7%	$7.4M	Mature MFI benchmarks
Base case	8%	$16M	New platform, limited history
Stress	15%	$30M	Thin credit history, volatile collateral

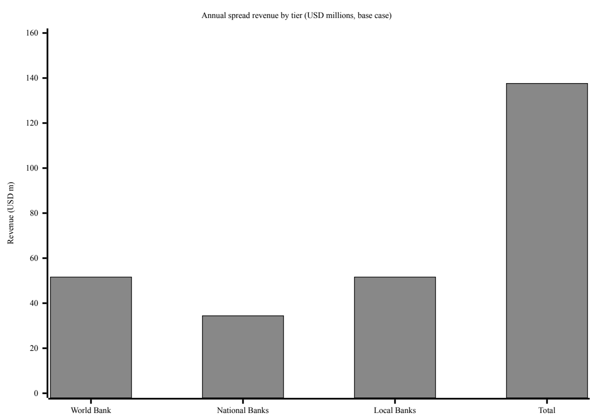


Figure 5.1: Annual revenue projection (10-year maturity)

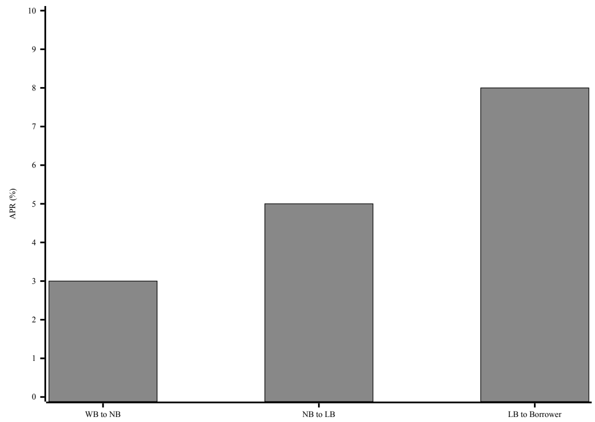


Figure 5.2: Hierarchical interest-rate spread (APR)

5.6 Comparisons and Relationships

Table 5.8: Market segments with supporting data

Segment	Description	Estimated Scale
TAM	Global DeFi lending; remittances; MSME gap [8, 11, 16]	$55B – 5T+
SAM	Institutional hierarchical lending; emerging-market credit	$5B – 15B
SOM	Testnet pilots; sandbox partnerships	$50 – 200M

Table 5.9: Target customer segment profile

Characteristic	Description
Primary users	Tier 1–3 bank operators; Tier 4 retail via Local Banks
Loan range	Micro/SME at Local Bank tier; larger exposures via syndication
Key pain points	Opaque reserves; flat DeFi pools; weak explainability in on-chain lending

Table 5.10: Partner categories and roles

Partner Category	Role	Incentive
Financial regulators	Sandbox oversight	On-chain audit trails
Banking institutions	National/Local Bank nodes	Reserve access; lower settlement friction
Oracle providers	Chainlink DON, indexing	Unified oracle stack
Academic partners	ML / formal-verification review	Datasets; co-publication

5.7 Discussion

This pre-thesis only shows architecture and evaluation plans. The deployment result, trained model and formal proof will be evaluated on pre-thesis 2. Five key limitations are:

- **Novelty is compound, not absolute:** this thesis uniquely represents the combination of a four-tier reserve system, explainable oracle ML and group liability lending within a single open framework while Maple, Goldfinch, and Morpho cover parts of institutional credit.
- **BCCC is a baseline only:** without testnet retraining, flat-pool fraud labels will not be transferred to hierarchical CWB flows.
- **Testnet scope:** to be accepted as strong research, the project needs performance tests with other systems and publicly available reports which people can run and verify. MVT deployment will help this real-world evidence.
- **Group lending:** BRAC and Grameen Bank statistical data is only used for designing purposes. The digital loan system or on-chain cosigning on the blockchain is not fully replaceable like real social support by humans.
- **Oracle and economics**—when credit scores are assigned, there are still chances of collusion or AI might drift. The Certora proof clearly specifies the assumptions and states that it might not give correctness under every possible scenario.

The prototype only works on testing environments. It is not ready for global release. It will be in the future.

Chapter 6

Conclusion

The *Crypto World Bank* addresses the coordination and trust challenges of traditional development finance through a decentralized four-tier lending architecture. Unlike existing DeFi lending protocols, the proposed system supports hierarchical capital flows between institutions, banks, lenders, and borrowers while enforcing reserve requirements at every tier.

The thesis introduces four key contributions:

- A four-tier DeFi lending architecture.
- Programmable group lending with over-indebtedness controls.
- Explainable AI-based fraud detection and credit scoring.
- Privacy-preserving identity and compliance using decentralized identifiers and zero-knowledge proofs.

Analysis of existing DeFi and banking solutions indicates that no current platform combines hierarchical lending, interbank operations, and tiered borrower access within a single decentralized system. The project provides a complete architectural design, governance framework, and implementation roadmap for a blockchain-based development-finance ecosystem.

6.1 Future Work

Complete end-to-end hierarchical lending, including cross-tier fund transfers, repayment mechanisms, and interest accrual. Deploy stablecoin-based financial services and develop savings, deposit, and lending products. Also integrate explainable AI analytics into live loan approval and fraud detection workflows.

We want to expand the platform with remittance services, merchant payments, fiat on/off ramps, and consumer protection features.

Lastly, for more advanced sections, this project can explore advanced research areas such as graph-based fraud detection, federated learning, privacy-preserving compliance, and formal verification of financial operations.

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